

A green lifetime biosensor for calcium that remains bright over its full dynamic range

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Franka H van der Linden, Stephen C Thornquist, Rick M ter Beek, Jelle Y Huijts, Mark A Hink, Theodorus WJ Gadella Jr., Gaby Maimon, Joachim Goedhart 

Swammerdam Institute for Life Sciences, Section of Molecular Cytology, van Leeuwenhoek Centre for Advanced Microscopy, University of Amsterdam, Amsterdam, Netherlands • Laboratory of Integrative Brain Function and Howard Hughes Medical Institute, The Rockefeller University, New York, United States

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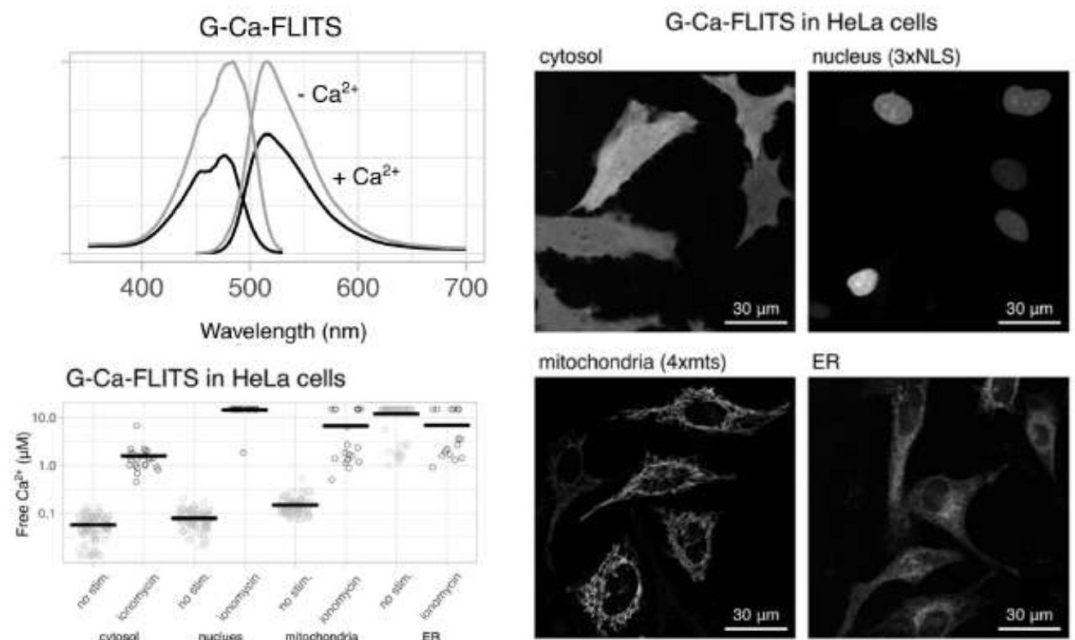
eLife Assessment

This manuscript reports on an FLIM-based calcium biosensor, G-CaFLITS. It represents an **important** contribution to the field of genetically-encoded fluorescent biosensors, and will serve as a practical tool for the FLIM imaging community. The paper provides **convincing** evidence of G-CaFLITS's photophysical properties and its advantages over previous biosensors such as Tq-Ca-FLITS. Although the benefits of G-Ca-FLITS over Tq-Ca-FLITS are limited by the relatively small wavelength shift, it presents some advantages in terms of compatibility with available instrumentation and brightness consistency.

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Abstract

Fluorescent biosensors toggle between two states and for the vast majority of biosensors one state is bright and the other state is dim. As a consequence, there is a substantial difference in the signal-to-noise ratio (SNR) for the two states. The dim state has a low signal-to-noise ratio, which is problematic when precise, quantitative measurements are needed. During the engineering of a red-shifted variant of an mTurquoise-based calcium sensor, we serendipitously generated a green-emitting sensor that shows high brightness in both the calcium-bound and -unbound state, while still showing a calcium dependent lifetime change of >1 nanosecond. This sensor, named G-Ca-FLITS, is comparable in brightness to the bright state of GCaMP3 and jGCaMP7c in mammalian cells. The calcium induced loss in fluorescence intensity is only around 30% and therefore we observe little variation in the SNR when calcium levels change. G-Ca-FLITS shows negligible sensitivity to pH in the physiological range, like its turquoise parent. Using fluorescence lifetime imaging (FLIM), we measured the calcium concentration with G-Ca-FLITS in various organelles and observed in HeLa cells transient and spatially heterogeneous calcium elevations in mitochondria. Finally, we evaluated the use of G-Ca-FLITS and its turquoise predecessor for two-photon FLIM in *Drosophila* brains.



Introduction

Genetically Encoded Calcium Indicators (GECIs) are by far the largest family of biosensors, with over 100 members (Greenwald et al., 2018) including the well-known GCaMP series. Many GECIs harboring a single fluorescent protein (FP), like GCaMPs, are optimized for a large intensity change, and have a (very) dim state when calcium levels are below the KD of the probe (Akerboom et al., 2013; Dana et al., 2019; Shen et al., 2018; Zhang et al., 2023; Zhao et al., 2011). This feature is advantageous when using the probe as a binary detector for increases in Ca^{2+} concentrations. However, it makes robust quantification of low, or even intermediate, calcium concentrations extremely challenging.

To enable more precise measurements of low calcium concentrations, a biosensor with an inverted response (Barykina et al., 2016) is a better alternative, as these biosensors are bright in the calcium-free state. However, this only partially addresses the issue, as the measurement precision for high calcium concentrations is compromised since the intensity is low in the calcium saturated state.

Another disadvantage for quantification with intensity-based biosensors is that the fluorescence intensity is affected by many factors (Culley et al., 2024; Waters, 2009), complicating the measurement of absolute calcium concentrations with biosensors that use intensity contrast as a read-out. Therefore, we and others have engineered genetically encoded biosensors that use other read-outs. New biosensors that report with a change in photochromic behaviour (Bierbuesse et al., 2022), anisotropy (Laskaratou et al., 2021), or fluorescence lifetime (Díaz-García et al., 2017; Klarenbeek et al., 2015; van der Linden et al., 2021) have been developed.

Despite the more robust read-outs, the measurement precision of all approaches depends on the number of detected photons. For instance, the turquoise lifetime sensor Tq-Ca-FLITS that we developed (van der Linden et al., 2021) shows a 3-fold intensity change. Yet, the ideal lifetime biosensor would be bright in both states, enabling high measurement precision across its entire dynamic range. Such a biosensor could be used for imaging low concentrations of calcium and subsequent increases, for example in organelles of mammalian cells, including mitochondria (Parry et al., 2024).

Calcium signaling in mitochondria is reported to be linked with several pathophysiological processes including inflammatory responses, diabetes, heart failure, neurological diseases and cancer (Delierneux et al., 2020 [DOI](#); Giorgi et al., 2018 [DOI](#)). Over the years, mitochondrial calcium concentrations have been measured with imaging techniques; however, there are discrepancies between the reported concentrations. Accurate determination of the affinity of a sensor is important and there are several issues that need to be considered during the calibration and the measurements: (i) the concentrations can only be measured with sufficient precision when it is in the range between $10 \times K_D$ and $1/10 \times K_D$, (ii) the calibration is only valid when the two extremes are reached during the calibration procedure (Fernandez-Sanz et al., 2019 [DOI](#)), (iii) the sensor's kinetics should be sufficiently fast enough to be able to track the calcium changes, and (iv) the biosensor should be compatible with the high mitochondrial pH of 8 (Cano Abad et al., 2004 [DOI](#); Llopis et al., 1998 [DOI](#)).

Clearly, there is a need for robust biosensors that satisfy all criteria discussed above, enabling quantitative imaging of intracellular calcium concentrations. In this paper we document the development of a green calcium sensor by an optimized bacterial screening method. The biosensor displays a strong lifetime contrast and at the same time remains bright in the calcium-free and -saturated states. We report the spectroscopic properties of the purified sensor and show its use for robust measurements of calcium levels in mitochondria of HeLa cells and endothelial cells by lifetime imaging.

Results

A high throughput method to screen sensors in bacterial lysate

Before creating sensor variants, we developed a high-throughput method to screen different calcium sensors in bacterial lysate. The aim was to grow and isolate 96 bacterial cultures in parallel in a deep well plate. A previously developed lysis buffer (Danilevich et al., 2008 [DOI](#)) served as a starting point for optimization of the protocol. Bacteria expressing the sensor Tq-Ca-FLITS were used to compare the lysis efficiency. The following variables were tested in a 50 mM Tris-HCl buffer: 1) presence or absence of a freeze/thawing cycle, 2) presence or absence of lysozyme, 3) different concentrations of urea (0, 0.5 and 1 M), 4) different concentrations of deoxycholic acid (DOC, 1, 2 and 3%). After lysis, the fluorescence intensity of the bacterial lysates was compared (Figure S1).

In general, a freezing and thawing cycle resulted in a higher fluorescence intensity for most conditions. However, this doubled the duration of the protocol. A high concentration of DOC and urea combined was too harsh for the fluorescent sensor: no intensity was detected. All other conditions were comparable. We favored a buffer without urea to maintain the native state of the proteins, and without lysozyme to omit the need to freshly add this component before each use. Therefore, we chose a protocol with 2% DOC and without a freezing step. The obtained lysates were of sufficient brightness and purity to use for spectral measurements and lifetime measurements.

We explored the possibility of replacing the 2% DOC with other detergents, but no improvements were found. Triton-X100 (0.1-1%) gave lower intensities compared to DOC and caused a problematic amount of foam during handling. Sodium dodecyl sulfate (SDS, 0.1-0.5%), an anionic surfactant in contrast to non-ionic Triton-X100 and DOC, also gave lower intensities and poses the risk of denaturation of the proteins (Figure S2). Therefore, we chose the 50 mM Tris-HCl buffer with 2% DOC for further experiments.

Creating a red shifted calcium sensor with lifetime contrast

We aimed to generate a green biosensor with lifetime contrast. Therefore, we used the previously developed turquoise sensor (Tq-Ca-FLITS) and mutated residues that potentially induce a red-shift. We subjected the Tq-Ca-FLITS to directed mutagenesis of residues T203 (mTurquoise2 numbering, T81 in the full sensor) and I167 (mTurquoise2 numbering, I45 in the full sensor) to aromatic amino acids. Both residues are located close to the chromophore. The mutation T203Y is known to cause a red shift in green fluorescent proteins (Ormö et al., 1996) and mTurquoise2 (Gorbachev et al., 2017). We anticipated that π - π stacking of the aromatic residue with the chromophore would result in the color shift in Tq-Ca-FLITS as well. The candidate sensors were expressed in bacteria and the bacteria were subjected to the optimized bacterial lysis protocol. The lysates were screened for a lifetime change and a change in excitation and emission spectra, measured in both the calcium-bound and calcium-free states by addition of CaCl_2 or EDTA (Table 1, Figure S3).

Mutations T203Y and T203F both resulted in a red-shifted excitation and emission spectrum in both calcium states, but the shift was more pronounced for the T203Y mutation. This Tq-Ca-FLITS_T203Y variant also showed a large phase lifetime contrast of $\Delta\tau_\phi = -0.9$ ns. Surprisingly, this change is inverted compared to Tq-Ca-FLITS ($\Delta\tau_\phi = 1.1$ ns). The T203F mutation also inverted the lifetime response; however, the contrast was about half of the Tq-Ca-FLITS_T203Y variant. A mutation of I167F or I167H in the background of the T203Y variant, or a separate I167F mutation on Tq-Ca-FLITS reduced the lifetime contrast and did not result in a (further) red shift.

We also tested the T203Y and T203H mutations on different circular permutations (cp) of Tq-Ca-FLITS, which we created in during the development of the turquoise sensor (van der Linden et al., 2021). In almost all cp variants with the T203Y mutation, the same red shift in both the excitation and emission spectra was observed as for Tq-Ca-FLITS_T203Y. An exception was the cp150-T203Y variant, where we observed a red shift only for the calcium-bound state in both excitation and emission spectra (Table S1, Figure S3). This variant differs from Tq-Ca-FLITS_T203Y by only a F146Y mutation. A T203H mutation did not cause a red shift in any of the cp variants.

Improving the green lifetime sensor by mutagenesis

We aimed to further improve Tq-Ca-FLITS_T203Y by mutagenesis in the regions linking the calcium binding domains to the FP. Using a PCR approach, we created a library of sensors with any possible amino acid on positions V27 and N271 (numbering according to the sensor, see Figure S4). Bright green fluorescent colonies on an agar plate were selected for the bacterial lysis test. We found three sensors with an improved lifetime contrast: with alanine or valine on position 27, and aspartic acid or serine at position 271 (Table 2). The sensor with V27A and N271D gave the best results with a contrast of $\Delta\tau_\phi = -1.4$ ns. Notably, the lifetime contrast of Tq-Ca-FLITS_T203Y is larger than measured in the previous screen (Table 1), which is due to a difference in temperature: the first screen was performed at 37 °C while this second one was performed at room temperature.

Next, we measured the lifetime response of the three improved variants in HeLa cells (Table 2), before and after saturation of the sensor with ionomycin and calcium. Again, the variant with V27A and N271D gave the best results, with a lifetime change of -1.2 ns (both $\Delta\tau_\phi$ and $\Delta\tau_M$, high calcium minus low calcium), which is an improvement of ~ 0.3 ns in lifetime change with respect to the parental sensor Tq-Ca-FLITS. We decided to name the new lifetime sensor G-Ca-FLITS, for Green Calcium Fluorescence LifeTime Sensor.

Characterization of G-Ca-FLITS

The G-Ca-FLITS has three mutations relative to its parent Tq-Ca-FLITS (Figure S4). In order to characterize its biophysical properties, we isolated G-Ca-FLITS from bacteria and purified the protein by a Ni-NTA column. The intermediate sensor Tq-Ca-FLITS_T203Y and a mTurquoise2 with

Table 1.

Lifetime and spectral screen of candidate sensors to create G-Ca-FLITS.

Sensor variants are indicated by their mutations with respect to Tq-Ca-FLITS. The calcium-bound state (sat) is measured in presence of 0.1 mM CaCl₂ and the calcium-free state (apo) after addition of 9.5 mM EDTA. *Emission and excitation maxima are only indicated if a red shift of the spectrum with respect to Tq-Ca-FLITS was observed. See also Figure S3. **Phase lifetimes were measured at 37 °C by FD-FLIM. The lifetime change is calculated as the lifetime in the calcium-bound state minus the calcium-free state.

Sensor variant	Excitation max. (nm)*		Emission max. (nm)*		Phase lifetime (ns) **		
	apo	sat	apo	sat	apo	sat	change
Tq-Ca-FLITS	437, 455	437, 456	492, 502	480, 507	1.72	2.86	1.14
T203F	459	458	506	507	1.71	2.19	0.48
T203H	no red shift	no red shift	no red shift	no red shift	2.82	2.90	0.08
T203Y	461, 488	459, 478	516	514	3.08	2.20	-0.88
I167F, T203Y	460, shoulder 480	460, 479	512, shoulder 490	516	2.59	2.88	0.29
I167H, T203Y	459, 478	460, 479	514	519	2.92	3.11	0.18
I167F	no red shift	no red shift	no red shift	no red shift	1.52	2.17	0.65

Table 2.

Lifetime contrast of improved green versions of Tq-Ca-FLITS.

*'AD' stands for the mutations V27A and N271D, 'VS' for V27 and N271S, 'AS' for V27A and N271S. **The fluorescence lifetime of bacterial lysates was measured at RT with by FD-FLIM after addition of 0.1 mM CaCl₂ and after addition of 9.5 mM EDTA. Mean phase lifetimes of the full field of view of the microscope are indicated. ***The fluorescence lifetime of HeLa cells expressing the different sensor variants was measured without stimulation and after addition of 5 µg/mL ionomycin and 5 mM calcium to the medium, measured at 37 °C. The mean and standard deviation (sd) of the fluorescence lifetime of all cells (n) are indicated.

Sensor variant*	State	Bacterial lysate**		Response in HeLa cells***						
		Phase lifetime (ns)		Phase lifetime (ns)			Modulation lifetime (ns)			
		mean	change	mean	sd	change	mean	sd	change	n
G-Ca-FLITS (Tq-Ca-FLITS_T203Y-AD)	apo	3.83	-1.37	3.19	0.08	-1.17	3.65	0.07	-1.15	314
	sat	2.45		2.02	0.06		2.50	0.08		267
Tq-Ca-FLITS_T203Y-VS	apo	3.39	-1.23	2.90	0.06	-0.92	3.37	0.07	-0.95	150
	sat	2.15		1.98	0.07		2.42	0.08		116
Tq-Ca-FLITS_T203Y-AS	apo	3.85	-1.23	3.19	0.08	-0.96	3.63	0.07	-0.89	306
	sat	2.62		2.23	0.06		2.74	0.06		345
Tq-Ca-FLITS_T203Y	apo	3.34	-1.13	2.95	0.08	-0.82	3.42	0.08	-0.85	193
	sat	2.21		2.13	0.06		2.57	0.07		201

T203Y mutation (mTq2_T203Y) were also purified for comparison (**Table 3**, Figure S5). G-Ca-FLITS shows excitation and emission spectra that are evidently red-shifted when compared to the Turquoise variant (**Figure 1**).

We determined the quantum yield (QY) of G-Ca-FLITS to be 41% for the calcium-free state, and 26% for the calcium-bound state (**Table 3**, Figure S6). This difference of 15% is slightly higher than in the intermediate variant Tq-Ca-FLITS_T203Y (13%). Interestingly, the quantum yield of mTq2_T203Y is 24%, comparable to the calcium-bound state of G-Ca-FLITS.

For G-Ca-FLITS, the extinction coefficient is for both states around $30,000 \text{ M}^{-1} \text{ cm}^{-1}$, which is slightly higher than mTq2_T203Y (**Table 3**, Figure S7).

We measured the pH sensitivity of G-Ca-FLITS and mTq2_T203Y (**Figure 2**, Figure S8). In the calcium-free state the G-Ca-FLITS an increase in fluorescence lifetime in the range pH 4-6, followed by a stable lifetime above pH = 6. In contrast, in the calcium-bound state, the lifetime increases until pH = 5.5, followed by a drop and a stable lifetime above pH = 7. The mTq2_T203Y protein also shows a maximum of fluorescence lifetime at a low pH (pH = 4) followed by a drop and a stable lifetime above pH = 6.

The calcium affinity of the lifetime readout of G-Ca-FLITS was determined at 37°C. To this end, we measured the lifetime of the sensor in a range of calcium buffers using Frequency-Domain FLIM (FD-FLIM) and plotted this in a polar space (**Figure 3A**, Figure S9). The two extreme values (zero calcium and 39 μM free calcium) are located on different coordinates in the polar plot and all intermediate concentrations are located on a straight line between these two extremes. Based on the position in the polar plot, we determined the fraction of sensor in the calcium-bound state, while considering the intensity contribution of both states (**Figure 3B**).

The intensity contribution was determined based on the excitation and emission spectra, the extinction coefficient per wavelength, the QY and the properties of the microscope, i.e. the excitation/emission filters, the excitation source and the detector sensitivity profile (Supplemental Note 1, Figure S10). A calibration curve was fitted to the fraction of sensors in the calcium-bound state for each concentration, from which the calcium affinity was determined to be $K_D = 209 \text{ nM}$ (**Table 4**). Using the variation in the measurements of the two extremes, we determined that we can reliably measure calcium concentration with G-Ca-FLITS at 37 °C between 14 nM and 16 μM . The same calibration was also performed at RT (**Table 4**, Figure S9). Here, the calcium affinity was $K_D = 339 \text{ nM}$, higher compared to the calibration at 37°C. This is in line with the notion that binding strength generally increases with decreasing temperature.

Little change in intensity between the two states

During engineering and characterization of G-Ca-FLITS, we noted a low intensity contrast for the two extreme states. For a quantitative brightness analysis in cells, we co-expressed G-Ca-FLITS and the red fluorescent protein mScarlet-I in HeLa cells from a single plasmid, using a P2A sequence (Kim et al., 2011; Liu et al., 2017).

We determined the ratio of green fluorescence over red fluorescence, to mitigate the effects of transfection efficiency, timing and cell thickness. The measurement was performed on both unstimulated cells, with a low intracellular calcium level, and cells stimulated with ionomycin and extra calcium to raise the calcium concentration to saturation of the sensor (**Figure 4**). To normalize the GFP/RFP ratio, we used EGFP-mScarlet-I and a construct without a green FP. G-Ca-FLITS and all its intermediate variants showed a higher intensity in both states compared to mTq2_T203Y. We measured a 0.70-fold change in GFP/RFP ratio for G-Ca-FLITS between the calcium saturated cells and the unstimulated cells.

Table 3.

Photophysical properties of G-Ca-FLITS and intermediate variant.

Apo – calcium-free state (10 mM EGTA), sat – calcium-bound state (39 μ M free Ca^{2+}), ϵ - Extinction coefficient, QY – quantum yield with 95% confidence interval between curly brackets.

	state	λ_{abs} (nm)	λ_{em} (nm)	ϵ_{max} ($\text{M}^{-1}\text{cm}^{-1}$)	QY {CI} (%)
G-Ca-FLITS	apo	474	515	29300	41.1 {40.5 - 41.7}
	sat	476	517	30900	25.9 {25.4 - 26.5}
Tq-Ca-FLITS_T203Y	apo	470	516	28900	43.8 {43.2 - 44.3}
	sat	475	518	30900	30.7 {30.4 - 31.1}
mTq2_T203Y		471	514	27400	23.8 {23.5 - 24.2}

Figure 1.

The excitation and emission peaks of G-Ca-FLITS are red-shifted relative to Tq-Ca-FLITS.

(A) The excitation and emission spectra show in lighter lines the calcium-free state (10 mM EGTA) and the darker lines the calcium-bound state (39 μ M free Ca^{2+}). The Tq-Ca-FLITS is shown as a reference and was reported before (van der Linden et al., 2021). Spectra are normalized to the maximum of the calcium-free state for G-Ca-FLITS. (B) The photo shows the fluorescence from tubes of purified biosensors when excited with UV (312 nm).

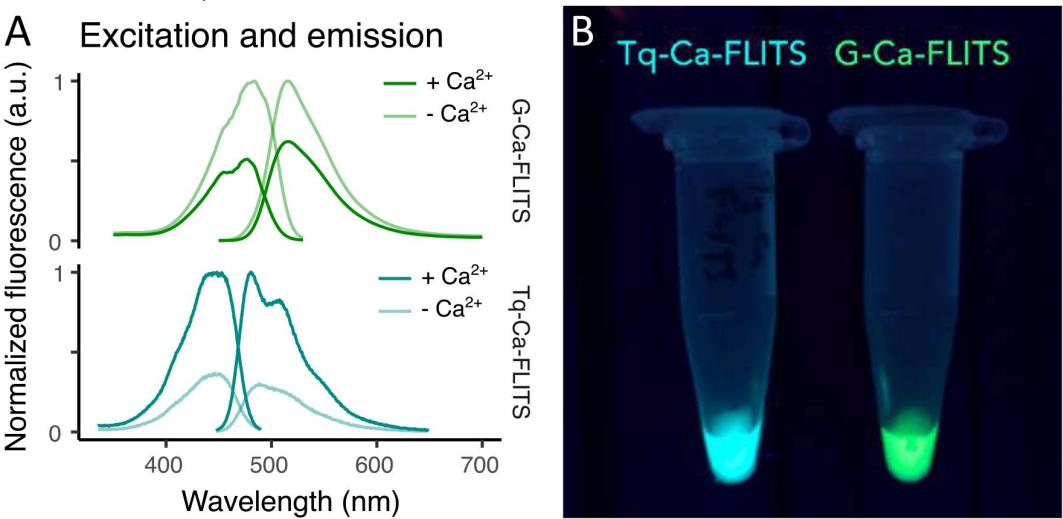


Figure 2.

Influence of pH on phase lifetime of the proteins.

Fluorescence lifetime of proteins diluted in pH buffer was measured ($n=3$). In case of G-Ca-FLITS, this was done in presence (gray, 0.1 mM CaCl_2) or absence of calcium (black, 5 mM EGTA). A smooth curve is fitted through the data using the loess method, using $\alpha = 0.4$. The gray band indicates the 95% confidence interval of the smooth fit.

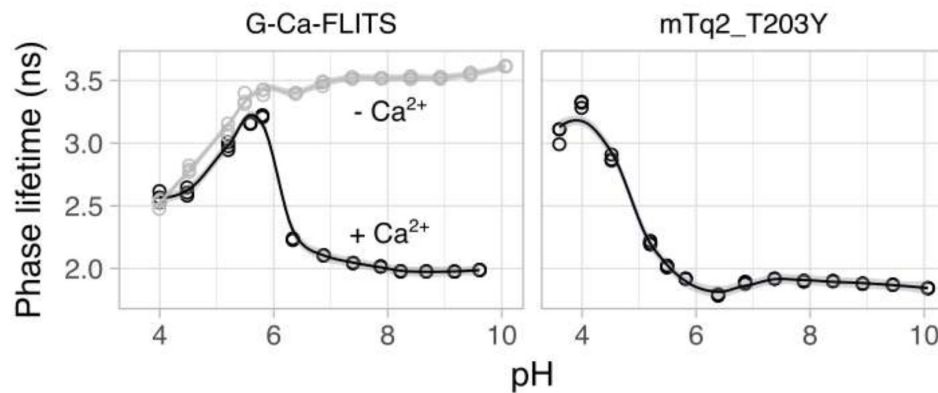


Figure 3.

Calcium calibration of G-Ca-FLITS in vitro at 37°C.

(A) The measured fluorescence lifetime of protein isolate of G-Ca-FLITS in a range of calcium concentrations is plotted in a polar space ($n=3$), with the color indicating the concentration. The measurements fall on a straight line (in gray) on the polar plot between the lowest and highest concentration (indicated by an X). (B) For each measurement, the fraction of sensors in the calcium-bound state (the low lifetime) is determined, taking the intensity contribution of the two extreme states into account. The fraction is plotted against the concentration of free calcium to obtain a calibration curve.

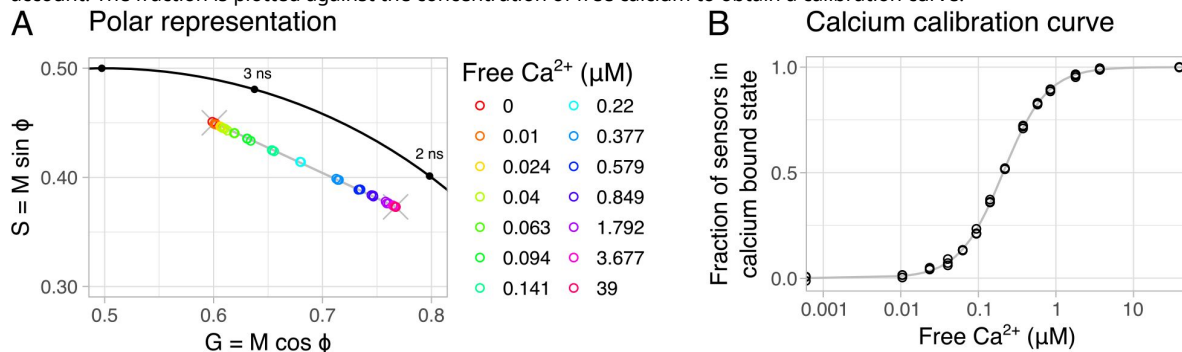


Table 4.

Calcium calibration parameters of G-Ca-FLITS.

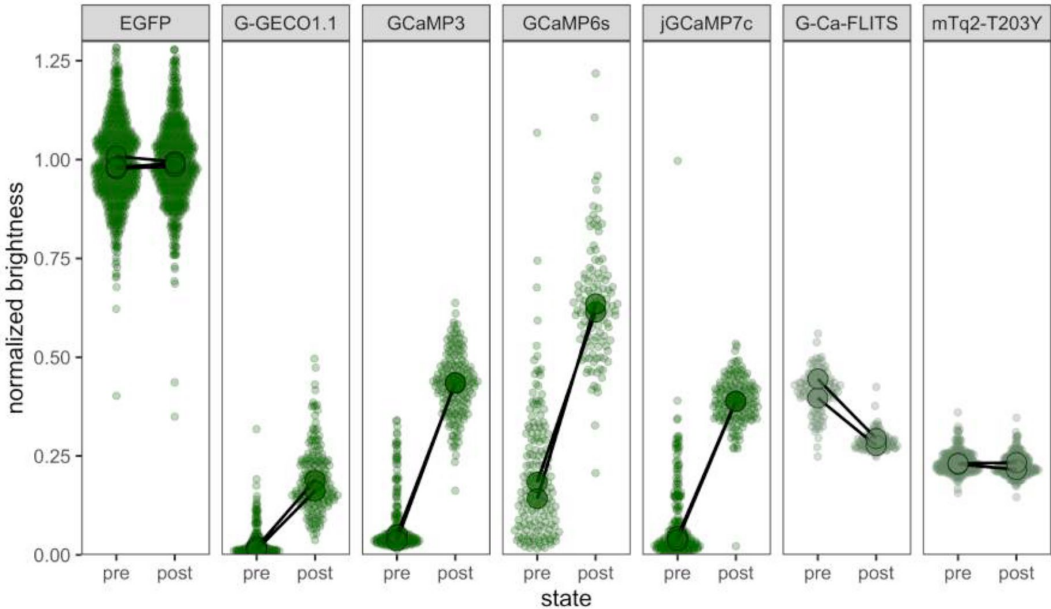
The calibration was performed with purified protein at two temperatures. The phase and modulation lifetime of the sensor are indicated for the two states of the sensor. apo – in calcium-free buffer (10 mM EGTA), sat – in buffer with 39 mM free CaCl_2 , τ_ϕ and τ_M – phase lifetime and modulation lifetime, K_D – calcium affinity with 95% confidence interval indicated between curly brackets, n – Hill coefficient with 95% confidence interval indicated between curly brackets.

Temperature	state	τ_ϕ (ns)	τ_M (ns)	K_D {CI} (nM)	n {CI}
37 °C	apo	2.98	3.50	209	1.53
	sat	1.93	2.43	{206-211}	{1.51-1.56}
RT	apo	3.52	3.99	339	1.67
	sat	2.03	2.57	{335-343}	{1.64-1.70}

Figure 4.

Brightness analysis of green fluorescent proteins and biosensors in HeLa cells using a co-expressed mScarlet-I.

The ratio GFP/RFP was determined 24 h after transfection both in resting, unstimulated cells (pre) and cell stimulated with 5 $\mu\text{g}/\text{mL}$ ionomycin and 5 mM calcium to reach saturation of the sensors (post). The ratios were normalized to cells expressing EGFP. Dots show individual cell data ($n \geq 99$) and the large dot the average of each biological replicate ($N = 2$, except for EGFP; $N=3$).



To compare G-Ca-FLITS to other green GECIs, we repeated the experiment with G-GECO and several variants of GCaMP (**Figure 4**). In contrast to G-Ca-FLITS, these sensors show a high GFP/RFP ratio in the calcium saturated state and a low ratio in resting cells, resulting in a large ratio difference between the two states. The fold change was 5.6, 3.1, 4.6 and 6.3 for GCaMP3, GCaMP6s, jGCaMP7c and G-GECO1.1 respectively. The relative intensity of the GCaMPs/GECO in resting cells is much lower compared to the dimmest state of G-Ca-FLITS. We also found a larger standard variation in GFP/RFP in resting cells expressing GCaMP/G-GECO compared to G-Ca-FLITS.

The precision of the fluorescence lifetime depends on the number of photons. We hypothesized that the fluorescence lifetime of G-Ca-FLITS would have a similar precision in both low and high concentrations of calcium, given the small change in intensity. To examine this, we determined the signal to noise ratio (SNR = mean / standard deviation) of the fluorescence lifetime of G-Ca-FLITS in cells, measuring an area of 400 pixels. The absolute SNR is not relevant here, as it will depend on the expression level and acquisition time. But since we have measured the two extremes in the same cells, we can evaluate how the SNR changes between these states for each separate probe (**Figure 5**, Figure S11). Indeed, the SNR of G-Ca-FLITS is on average only 13% lower for the high calcium state. In comparison, the SNR of Tq-Ca-FLITS increases more than 2-fold when going from the low to high calcium state, in agreement with its 3-fold change in intensity. In all cases, the variation of the average lifetime of all cells is similar, meaning the average lifetime in the 400 pixels could be properly determined.

Measurement of calcium in organelles

Several tags were added to G-Ca-FLITS to target different organelles: the cytosol, the nucleus, the mitochondrial matrix, and the endoplasmic reticulum (ER). With the bright fluorescence of the probe independent of calcium, we could easily confirm the correct localization of the sensor to all targeted organelles, in both HeLa cells and Blood Outgrowth Endothelial Cells (BOECs) (**Figure 6A**, **Figure 6B**).

Next, we measured the calcium concentration in the various organelles in the same two cell types, using widefield FD-FLIM. Without stimulation, HeLa cells showed an average calcium concentration of 57 nM in the cytosol, 79 nM in the nucleus and 148 nM in the mitochondria. After addition of 5 µg/mL ionomycin and 5 mM calcium, the average concentration was raised above 1 µM. In the ER the concentration was above 1 µM in all experiments. BOECs showed the same pattern, with average concentrations of 73 nM in the cytosol, 101 nM in the nucleus, 119 nM in the mitochondria and above 1 µM in the ER. After stimulation, the concentration in all organelles was elevated to above 1 µM (**Figure 6C**, **Figure 6D**).

In both cell types, no spatial differences were observed in unstimulated cells, except for the mitochondria of HeLa cells. Here, about 20% of the cells, had in peripheral parts a higher calcium concentration, comparable to the concentration in mitochondria after stimulation with ionomycin. These areas are not included in the calculations of the aforementioned averages.

HeLa cells expressing G-Ca-FLITS in the cytosol were stimulated with 1 µM histamine. We observed various patterns of oscillating calcium concentrations, confirming that binding of calcium to G-Ca-FLITS is reversible (**Figure 6E**). The calcium concentration was elevated to about 1 µM after addition of ionomycin and calcium, as seen before. The same experiment was repeated with HeLa cells expressing G-Ca-FLITS in the mitochondria (**Figure 6F**). Stimulation with histamine resulted in either no reaction or an elevation of the calcium concentration followed by a drop to basal level. Addition of ionomycin and calcium resulted in high concentrations as expected.

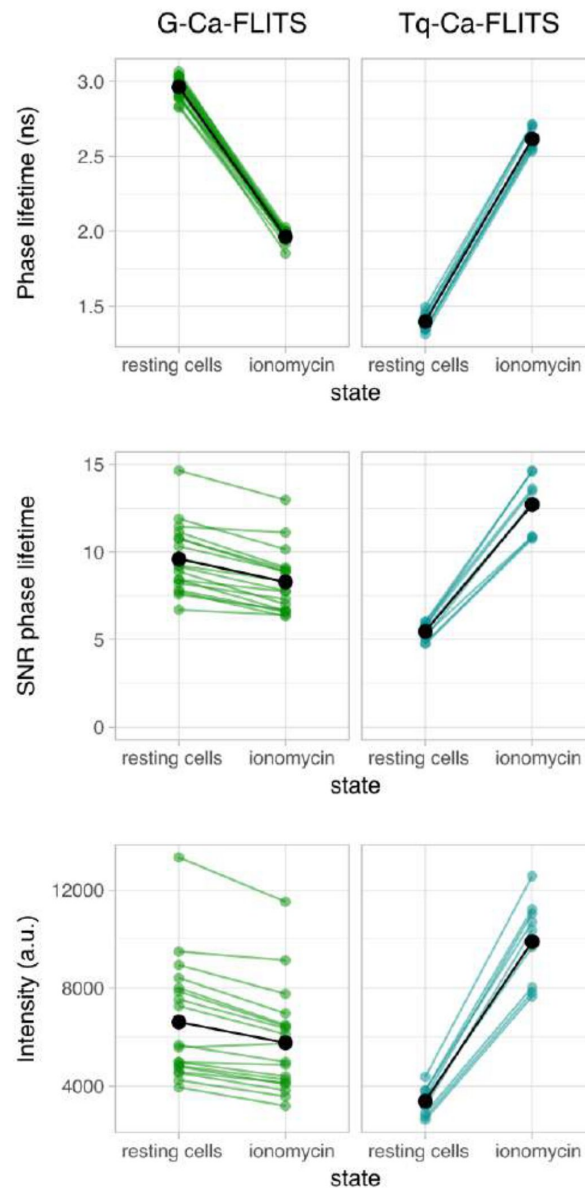


Figure 5.

Comparison of the signal-to-noise ratio of the phase lifetime of G-Ca-FLITS and Tq-Ca-FLITS in HeLa cells.

HeLa cells were measured in a resting state and after addition of 5 $\mu\text{g/mL}$ ionomycin and 5 mM calcium. Measurements of individual cells are indicated by colored dots ($n=18$ for G-Ca-FLITS, $n=9$ for Tq-Ca-FLITS). Lines connect the measurements of the same cell. The average of all cells is shown in black. For each cell, 400 pixels were analyzed for the mean lifetime and intensity, and for the sd of the lifetime. $\text{SNR} = \text{mean lifetime} / \text{sd lifetime}$. Comparable phase lifetimes are found for all cells expressing a construct but the SNR is lower for lower intensity cells and varies less between the two states for G-Ca-FLITS.

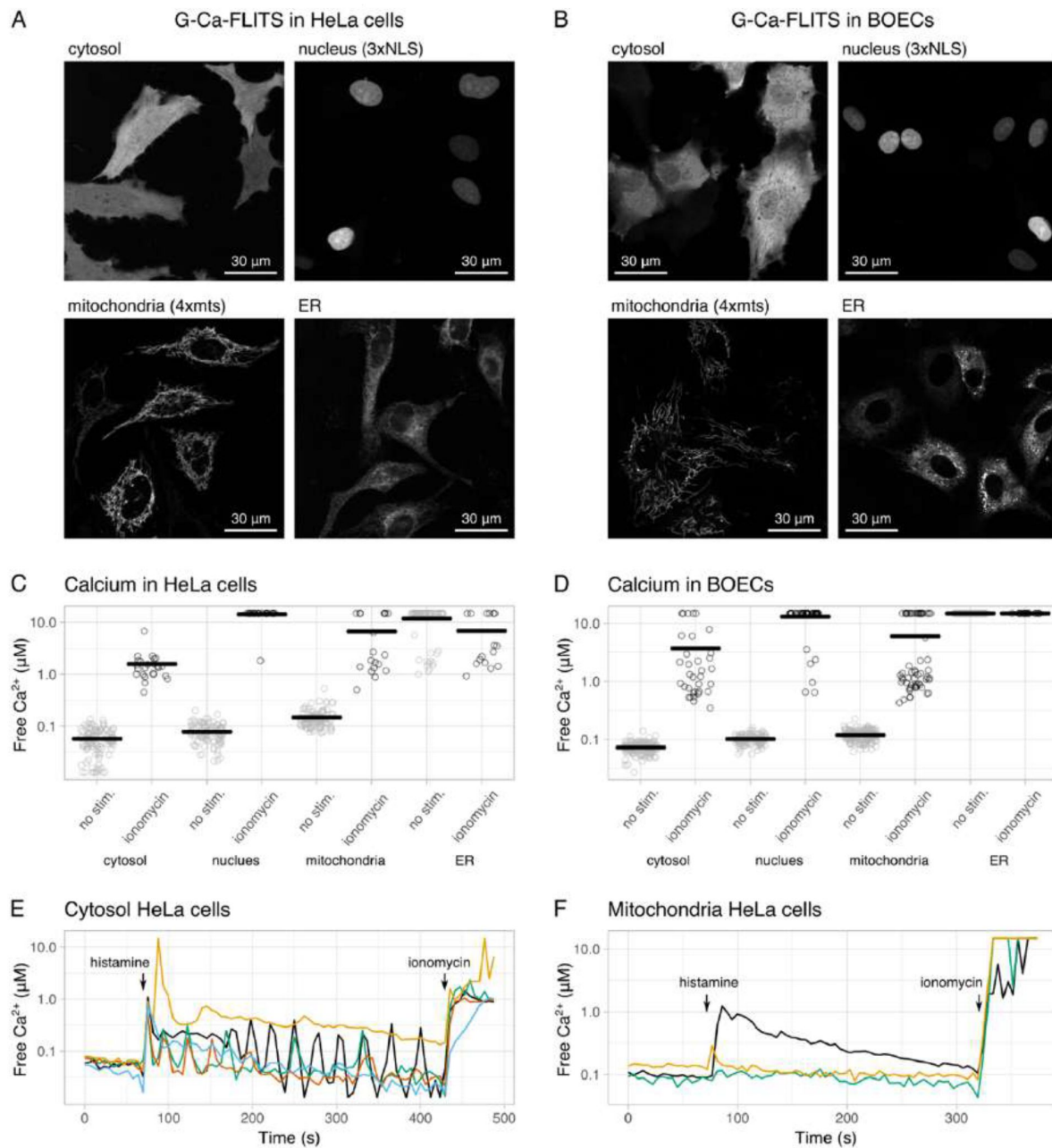


Figure 6.

Measuring calcium concentrations in HeLa cells and BOECs using G-Ca-FLITS targeted to various organelles.

(A) Localization of G-Ca-FLITS in HeLa cells and (B) in BOECs. Images are taken with a 63x magnification. (C) Measured calcium concentration in various organelles in HeLa cells and (D) BOECs. Measurements of single cells are indicated by circles, gray for unstimulated cells and black for cells after addition of 5 $\mu\text{g}/\text{mL}$ ionomycin and 5 mM calcium. The mean of all cells is indicated by a black line. (E) Changing free calcium concentration after stimulation with 1 μM histamine or addition of 5 $\mu\text{g}/\text{mL}$ ionomycin and 5 mM calcium in the cytosol and (F) in the mitochondria. Each line represents a single cell. Arrows indicate the moments of addition.

Since confocal imaging of fluorescence lifetime results in images with a better spatial resolution, we used a confocal microscope with a Time Correlated Single Photon Counting (TCSPC) detector to examine the heterogeneity that we observed in mitochondria. From the confocal lifetime images, it is evident that several cells have mitochondria that show a decreased lifetime in the periphery of the cell (Figure S12). This indicates that mitochondria show heterogeneity and that a pool of mitochondria has increased calcium levels.

The confocal TCSPC system needs relatively long integration times (several minutes) and we turned to a Leica Stellaris system which allows for a >10-fold higher temporal resolution to study calcium dynamics. We introduced the 4mts-G-Ca-FLITS into HeLa cells and acquired a timelapse of lifetime images. At the end of the experiment digitonin was added to acquire the minimal lifetime, reflecting maximal calcium. Using the calibration curve that was generated using purified protein incubated with different calcium buffers (Figure S13) we transformed the FLIM data into calcium concentration images as we did before. The calcium concentration images show the heterogeneity in calcium concentration and its dynamics over time (Figure 7). The elevated calcium concentrations are reversible but can last several minutes. Intriguingly, there is substantial difference in the dynamics between patches of mitochondria.

Multiplex imaging

The green FLITS probe has a red-shift of approximately 35 nm in both excitation and emission peaks relative to its Turquoise parent, resulting in a strikingly different emission color (Figure 1). Therefore, we examined whether we can use these probes together in a single cell for multiplex imaging. To this end, we co-expressed a nuclear targeted 3xnl::Tq-Ca-FLITS and a G-Ca-FLITS-3xnes with a triple nuclear export sequence (nes). We selected an excitation wavelength and an emission bandpass to selectively image either of the two probes and, as a result, we could distinguish the differential localization of the two probes (Figure 8). Upon stimulation of the cells with histamine, we observed a lifetime change for both probes, with the lifetime of Tq-Ca-FLITS increasing and the lifetime of G-Ca-FLITS decreasing as expected, since histamine increases the intracellular calcium concentrations.

2P-FLIM

To assess the functionality of G-Ca-FLITS and Tq-Ca-FLITS with two-photon time-correlated single-photon counting (TCSPC) FLIM microscopy, we measured calcium in the brains of *Drosophila melanogaster* individuals affixed to a previously described stage (Maimon et al., 2010). Studies of activity-dependent transcription in neurons have used saline with elevated $[K^+]$ to induce sustained and controlled increases in intracellular calcium. We employed a similar approach to raise external $[K^+]$ from 3 mM to 103 mM by perfusing saline over the brain while imaging jGCaMP7f (Dana et al., 2019), Tq-Ca-FLITS and G-Ca-FLITS signals. Our experiments here focus on the EPG neurons, a population of neurons innervating the ellipsoid body region of the fly brain and which serve a role in navigation-related computations (Green et al., 2019; Seelig and Jayaraman, 2015) (Figure 9A).

jGCaMP7f exhibited large changes in fluorescence intensity several minutes after we elevated external $[K^+]$ (Figure 9B). After a number of minutes (with the exact number varying from fly to fly), calcium levels fluctuated between low and high states synchronously across the field of view (including the EPG neurites visible in the middle of Figure 9E and the neurites of off-target cells in the bottom corners of the same images). The sample with jGCaMP7f showed only a very small change in lifetime in response to elevated calcium (0.030 nanoseconds ± 0.003 , mean $\pm$ sd) despite the many-fold shift in fluorescence intensity ($5.79 \Delta F/F \pm 0.503$ (mean $\pm$ sd)).

Figure 7.

Quantitative imaging of spontaneous calcium dynamics in mitochondria.

(A, B) FLIM images were acquired with a Leica Stellaris8 every 60 seconds and at the end of the sequence 10 μM digitonin was added to obtain a maximal response. The lifetime images were converted to calcium concentration using the K_D determined in vitro and the extremes from the lifetime image. False colors reflect the calcium concentration, according to the scale bar on the left. Calcium image at $t=1$ (A) and $t=11$ minutes (B), showing the three regions where calcium concentrations were quantified and displayed over time (C).

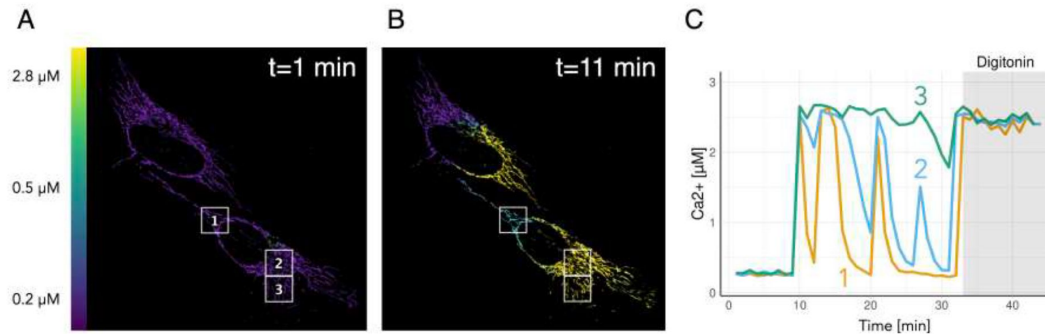
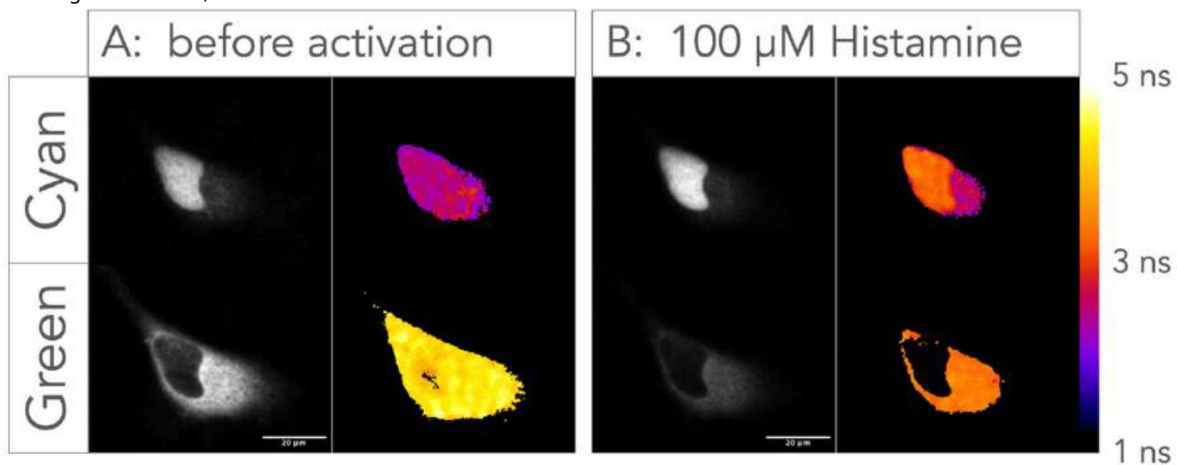


Figure 8.

Multiplex imaging of Tq-Ca-FLITS and G-Ca-FLITS.

(A, B) FLIM images of HeLa cells co-transfected with 3xnlis-Tq-Ca-FLITS and G-Ca-FLITS-3xnes were acquired with a Leica Stellaris8 before (A) and after addition of 100 μM histamine (B). False colors reflect the lifetime (average arrival time), according to the scale bar on the right. For the cyan channel, the excitation was at 440nm and detection at 450nm-490nm and for the green channel, the excitation was at 490nm and detection at 500nm-540nm.



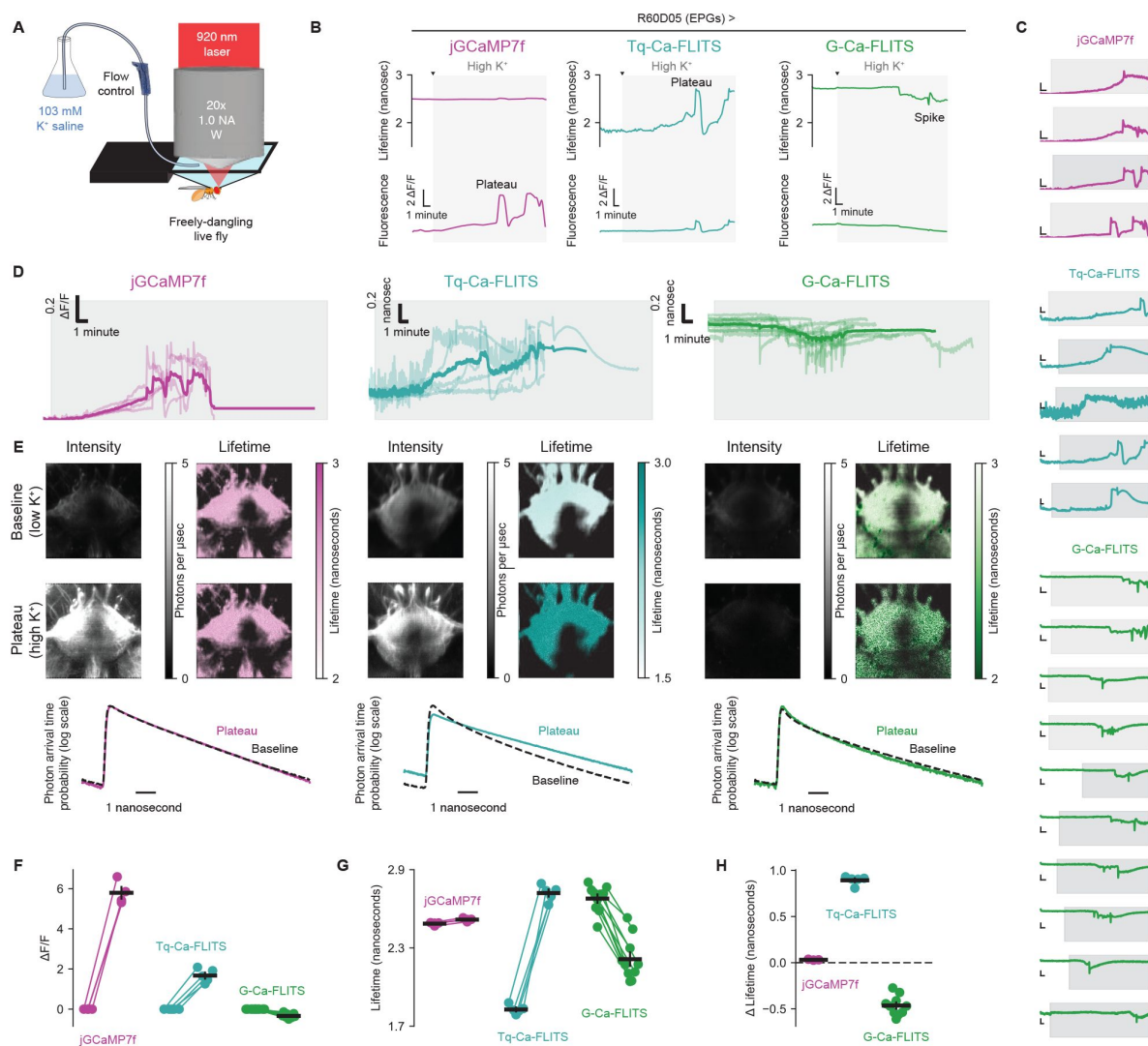


Figure 9.

Two-photon TCSPC signals from G-Ca-FLITS and Tq-Ca-FLITS, and comparison to jRCaMP7f, in an intact *Drosophila* brain.

(A) Schematic of a head-fixed live *Drosophila melanogaster* female, imaged with standard saline for two minutes before perfusing a high $[K^+]$ saline variant over the brain. (B) Single example traces of flies expressing jRCaMP7f (pink, left), Tq-Ca-FLITS (turquoise, center) or G-Ca-FLITS (green, right) in the EPG neurons under the control of R60D05-Gal4. Top row: Tq-Ca-FLITS and G-Ca-FLITS show strong FLIM changes in response to elevated $[K^+]$ while jRCaMP7f shows very little. Bottom row: jRCaMP7f and Tq-Ca-FLITS exhibit large changes in fluorescence while G-Ca-FLITS becomes moderately dimmer. Gray region indicates time when high K^+ saline was perfused. (C) Intensity traces from all flies. Vertical scalebars: 0.200 nanoseconds for G-Ca-FLITS and Tq-Ca-FLITS, 5 $\Delta F/F$ for jRCaMP7f. Horizontal scalebars: 1 minute. Gray region indicates time when high K^+ saline was perfused. (D) Aggregated responses, the thick lines represent the mean response of all measurements. (E) Fluorescence (left) and FLIM (right) images of example flies from panel B before $[K^+]$ elevation (baseline, 2 minute average, top row) and for the 30 second surrounding the peak change in calcium (second row). Lifetime images are masked to show only automatically-determined "foreground" pixels. Third row: photon arrival time histograms for the data of the above rows. (F-H) Summary statistics for all flies (from panel C). (F) shows changes in fluorescence intensity, (G) shows absolute lifetime measurements, and (H) reflects the change in lifetime from baseline to plateau for each of the three indicators.

Tq-Ca-FLITS fluorescence was higher than that of jGCaMP7f at baseline (**Figure 9E** [↗](#)). In contrast, despite the high brightness of G-Ca-FLITS in the apo state in other systems, the baseline intensity of G-Ca-FLITS in these fly neurons was much lower than both jGCaMP7f and Tq-Ca-FLITS, for unclear reasons. Even with increased laser power, G-Ca-FLITS was dimmer than jGCaMP7f at baseline (see Methods) (**Figure 9E** [↗](#), top row).

We evaluated the relative brightness of purified Tq-Ca-FLITS and G-Ca-FLITS on beads by either 1-Photon Excitation (1PE) (at 460 nm) or 2-Photon Excitation (2PE) (at 920 nm) and observed a similar brightness between the two modes of excitations (Figure S14). This shows that the two probes have similar efficiencies in 2PE and suggest that the low brightness of G-Ca-FLITS in *Drosophila* is due to lower expression or poor folding.

In response to elevated $[K^+]$, Tq-Ca-FLITS showed large increases in fluorescence ($1.672 \Delta F/F \pm 0.299$, mean $\pm$ sd) and empirical lifetime (0.893 nanoseconds ± 0.043 , mean $\pm$ sd) (**Figure 9F-G** [↗](#)). The response of G-Ca-FLITS was also consistent with increases in intracellular calcium: this sensor exhibited large decreases in empirical lifetime (-0.465 nanoseconds ± 0.101 (mean $\pm$ sd)), and a modest decrease in intensity ($-0.339 \Delta F/F \pm 0.043$, mean $\pm$ sd) after perfusion of elevated external $[K^+]$ (**Figure 9F-H** [↗](#)).

Overall, all three fluorophores exhibit calcium-dependent signals in an intact brain using 2P-FLIM. The Tq-Ca-FLITS lifetime response is particularly large and shows much higher brightness under 2P excitation than G-Ca-FLITS in both the baseline and high calcium states.

Discussion

Most fluorescent biosensors toggle between a dim and a bright state. The dim fluorescence decreases the SNR, thereby increasing the measurement error in quantitative imaging. Here, we present a green calcium biosensor, G-Ca-FLITS, that maintains a consistent brightness over its full dynamic range. The calcium-dependent lifetime decrease of 1.2 ns when calcium is elevated can be used to quantify calcium concentrations in cells. In comparison, the cyan/turquoise sensor Tq-Ca-FLITS we previously developed ([van der Linden et al., 2021](#) [↗](#)) shows a 3-fold change in intensity combined with a similar change in lifetime. The intensity of G-Ca-FLITS is higher than the dim (calcium-free) state of several popular green GCaMPs, and comparable to the bright state of most tested sensors, as determined by ratio imaging with co-expression of mScarlet-I. Of the tested GCaMPs, GCaMP6s seems to have a fairly bright calcium-free state, and we believe this is due to the higher calcium affinity ($K_D = 144$ nM) of this sensor ([Chen et al., 2013](#) [↗](#)). In contrast, the affinities are 339 nM, 298 nM and 618 nM at room temperature, for G-Ca-FLITS, jGCaMP7c ([Dana et al., 2019](#) [↗](#)) and G-GECO1.1 ([Zhao et al., 2011](#) [↗](#)), respectively.

The minimal change in fluorescence intensity compared to other commonly used calcium biosensors results in a similar SNR for both states of G-Ca-FLITS. Another advantage of a consistently bright probe is that cells and structures labeled with the biosensor can be readily visualized. Therefore, low concentrations of calcium can be precisely measured in unstimulated mammalian cells, as well as increased concentration after stimulation. The quantitative measurements require calibration of the biosensor response to different calcium concentrations. Although the calibration is equipment independent under ideal conditions, and theoretically only needs to be performed once, we do see differences between different set-ups. Therefore, it is necessary to repeat the calibration for different set-ups. We prefer to use the phasor analysis (which can be used for both frequency- and time-domain FLIM), as it makes no assumptions about the underlying decay kinetics.

Our measured concentrations in resting cells agree with literature, 20 - 100 nM is reported for cytoplasm and nucleus, and a slightly higher concentration of 50 - 300 nM in the mitochondria (Arnaudeau et al., 2001 [↗](#); Berridge et al., 2000 [↗](#); Dalal et al., 2020 [↗](#); Palmer et al., 2006 [↗](#); Rotrosen and Gallin, 1986 [↗](#); Worthen and Nollert, 2000 [↗](#)).

We note that the underlying assumption of the quantification of organellar calcium concentrations is that the lifetime contrast is the same in each organelle. This is broadly true for most of the measurements (Figure S15). Yet, there are also differences. It is currently unclear whether the discrepancies are due to differences in the physicochemical properties of the compartments, or whether there is a technical reason (the efficiency of ionomycin for saturating the biosensor in the different compartments is unknown, as far as we know). This is something that is worth revisiting. A related issue that deserves attention is the level of agreement between in vitro and in vivo calibrations.

All calcium indicators also act as buffers, and this limits the accuracy of the absolute measurements, especially for the lower calcium concentrations (Rose et al., 2014 [↗](#)), as the expression of the biosensor is usually in the low micromolar range. Free calcium in the ER is reported to be >300 μ M, well above saturation of G-Ca-FLITS (Arnaudeau et al., 2001 [↗](#)).

Using G-Ca-FLITS, we were able to detect various (oscillating) patterns of calcium in HeLa cells after stimulation with histamine. These data show that the sensor is suitable for detection of repeated transient calcium elevations. Importantly, G-Ca-FLITS gives a stable read-out at the high pH of 8.0 in mitochondria. We observed high calcium level in some peripheral mitochondria in HeLa cells, but not in endothelial cells. Others have shown heterogenous responses of calcium (Greotti et al., 2019 [↗](#); Palmer et al., 2006 [↗](#)) and electrical signals (Collins et al., 2002 [↗](#)) in mitochondria. However, this is different from the sustained high levels we observed in exclusively peripheral parts of the cells, without any stimulation. Studies show that mitochondria can be distinguished in different pools based on their location and function (Ngo et al., 2021 [↗](#); Park et al., 2001 [↗](#)). Peripheral mitochondria are one of these groups and have less connections to the ER compared to other pools. These connections play an important role in calcium signaling of the mitochondria (Giorgi et al., 2018 [↗](#)). In cancerous cells, like HeLa cells, the mitochondrial calcium homeostasis can be dysregulated (Delierneux et al., 2020 [↗](#)). We speculate that this leads to the sustained high levels of calcium in some peripheral mitochondria, possibly linked to aberrant connections to the ER.

To our knowledge, the sustained high levels we observed in mitochondria have not been reported yet. The types of probes used so far might be a contributing factor. Detecting the sustained high levels without stimulation is hardly possible with a probe that only shows an intensity response, like the widely used Rhod-2 or Aequorin. It is possible by using filterFRET/sensitized emission (Gadella, 2009 [↗](#)), but this requires extensive controls. Also, much research in mitochondria has been focused on the rise in calcium after stimulation, and therefore probes with a high K_D (over 1 μ M) have been used preferably (Arnaudeau et al., 2001 [↗](#); Fernandez-Sanz et al., 2019 [↗](#); Fonteriz et al., 2010 [↗](#); Palmer et al., 2006 [↗](#)). This low affinity might be too low to detect the effect we have seen.

G-Ca-FLITS shares its >1 ns lifetime contrast and pH stability with the cyan/turquoise sensor Tq-Ca-FLITS we previously developed (van der Linden et al., 2021 [↗](#)), but the green emission is advantageous for several reasons: 1) excitation for green FPs is installed on virtually every lifetime microscope, while excitation for cyan FPs is less common, 2) the phototoxicity will be less and the penetration depth will be improved, and 3) it provides possibilities for multiplexing with Förster Resonance Energy Transfer (FRET)-FLIM sensors that often harbor a cyan-yellow FP pair, for example EPAC (Klarenbeek et al., 2015 [↗](#)).

Since biosensors with a CFP-like chromophore are largely pH insensitive (van der Linden et al., 2021 [↗](#); Zhong et al., 2024 [↗](#)), and we show here that the pH independence is retained for the Green Ca-FLITS, we expect that adding the T203Y mutation to a cyan sensor is a good approach for generating pH-insensitive green lifetime-based sensors.

The unique properties of the new green calcium sensor are evident from its photophysical properties. The extinction coefficient of the calcium-bound and -free states of G-Ca-FLITS is very similar, but the difference in QY is significant. Interestingly, the lower QY is observed in the calcium-bound state and is very similar to the QY of mTq2_T203Y, the FP that is used in this probe. This, combined with the resemblance in spectral shape and pH sensitivity, indicates that in the calcium-bound state the FP-barrel is probably in roughly the same closed conformation as mTq2_T203Y. When calcium is removed, a more open state is assumed with a more favorable QY, and a higher lifetime. Possibly, the lifetime contrast could be further improved by increasing the QY of the calcium-free state

Usually, a higher QY results in a higher intensity; however, in G-Ca-FLITS the open state has a differential shaped excitation spectrum which leads to a decreased excitation efficiency. These effects combined have resulted in a sensor where the two different states have a similar intensity despite displaying a large lifetime contrast. We note that this effect is dependent on the excitation wavelength. The smallest intensity change is obtained when the calcium-bound state is preferably excited (i.e. near 450 nm) and the effect is less pronounced when the probe is excited near its peak at 474 nm.

We evaluated the use of Tq-Ca-FLITS and G-Ca-FLITS for 2P-FLIM and observed a surprisingly low brightness of the green variant in an intact fly brain. This result is consistent with a study finding that red-shifted fluorescent-protein variants that are much brighter under one-photon excitation are, surprisingly, dimmer than their blue cousins in multi-photon microscopy (Molina et al., 2017 [↗](#)). The responses of both probes were in line with their properties in single photon FLIM, but given the low brightness of G-Ca-FLITS under 2-photon excitation in *Drosophila*, the Tq-Ca-FLITS is a better choice in this system. Yet, the brightness of G-Ca-FLITS with 2PE at 920 nm is comparable to Tq-Ca-FLITS, so we expect that 2P-FLIM with G-Ca-FLITS is possible in tissues that express it well.

In conclusion, we developed a green fluorescent GECI, named G-Ca-FLITS, with a lifetime contrast of over 1 ns, and a modest intensity change. This sensor does not have one (very) dim state unlike other green GECIs, therefore the variation in the measurement is comparable in both states. In addition, the lifetime response is stable above pH 7. We successfully measured the calcium concentration in various organelles in both HeLa cells and BOECs, including the mitochondria. A portion of HeLa cells show sustained high levels of calcium in some peripheral mitochondria, but further research is required to determine the origin of this heterogeneity. We anticipate that the G-Ca-FLITS will be a valuable tool to examine the heterogeneity of calcium concentrations in mitochondria and other cellular compartments.

Methods

General Cloning

We used *E. coli* strain *E. coli* 5-alpha (short: *E. coli*, Lucigen corporation) for all cloning procedures. For DNA assembly, competent *E. coli* was transformed using a heat shock protocol according to manufacturers' instructions. For protein expression, *E. coli* was grown using super optimal broth (SOB, 0.5% (w/v) yeast extract, 2% (w/v) tryptone, 10 mM NaCl, 20 mM MgSO₄, 2.5 mM KCl) supplemented with 100 µg/mL kanamycin and 0.2% (w/v) rhamnose (SKR). For agar

plates, 1.5% (w/v) agar was added. Bacteria were grown overnight at 37 °C. Plasmid DNA was extracted from bacteria using the GeneJET Plasmid Miniprep Kit (Thermo Fisher Scientific) and the obtained concentration was determined by Nanodrop (Life Technologies).

DNA fragments were generated by PCR, using Pfu DNA polymerase (Agilent Technologies) unless otherwise indicated. DNA fragments were visualized by gel electrophoresis on a 1% agarose gel, run for 30 min at 80 V. PCR fragments were purified using the GeneJET PCR purification Kit (Thermo Fisher Scientific) and digested with restriction enzymes to generate sticky ends. Restriction enzymes were heat inactivated at 80 °C for 20 min if necessary. Vector fragments were generated by restriction of plasmids and the correct bands were extracted from gel using the GeneJET Gel Extraction Kit (Thermo Fisher Scientific). DNA fragments were ligated using T4 DNA ligase (Thermo Fisher Scientific), per the manufacturers' protocol.

For targeted mutagenesis, we used the protocol as described before (Bindels et al., 2014). Primers were designed with an annealing region of 15 bp both left and right of the desired mutation(s). The full plasmid was amplified by PCR and the DNA was digested with DpnI to remove template DNA. *E. coli* was transformed with the digested PCR mix and individual colonies were selected for isolation of individual colonies.

Correct construction of plasmids was verified by control digestion and sequencing (Macrogen Europe). All primers (Table S2) were ordered from Integrated DNA Technologies.

Engineering of G-Ca-FLITS

Tq-Ca-FLITS was subjected to mutagenesis of residues T203 and I167 (both mq2 numbering), by cloning using in vitro assembly, IVA (García-Nafria et al., 2016). First, full plasmids were replicated by PCR while introducing the desired mutation using the dual expression plasmid pFHL-Tq-Ca-FLITS (Addgene plasmid #129628) as template, primers 1-2, 3-4 or 5-6 and Phusion DNA polymerase (Thermo Fisher Scientific). The resulting PCR mix was DpnI digested and 4 µl of the mix was used for transformation of *E. coli*. The linear DNA fragments were assembled into a plasmid in bacteria using a 15 bp homologous region at the ends of the fragments. In addition, intermediate, differently circular permuted variants of Tq-Ca-FLITS were subjected to mutagenesis of T203 by using primers 1-2 and the IVA method described above. Individual colonies were picked for testing of the excitation and emission spectra, for testing of the lifetime response and for plasmid isolation and sequencing.

The best performing variant was subjected to further mutagenesis of the sequences linking the FP with the CaM and M13 peptide. A library of variants was created by PCR with pFHL-Tq-Ca-FLITS as template and primers 7-8. The PCR fragments and a pFHL backbone (containing R-GECO1, a red calcium sensor) were digested using SacI and MluI, followed by ligation. Individual colonies were picked for screening of the excitation and emission spectra and of the lifetime response. Plasmids of interesting variants were isolated for sequencing.

Generation of the dual expression vector for ratiometric measurements

The SacI restriction site was removed from mScarlet-I in an 'empty' pDress plasmid (Bindels et al., 2020), containing mScarlet-I linked to an anti-FRET linker and a P2A site (Addgene plasmid #130509, but with the mScarlet protein between the AgeI and BsrGI sites replaced with the DNA sequence GCCTCAACATGA and mTurquoise2 replaced by mScarlet-I). The SacI site was removed by directed mutagenesis using primers 9-10, see '**General cloning**'. After mutagenesis, correct removal of the SacI site and the absence of unwanted spontaneous mutations was verified by control digestion with SacI and NheI followed by sequencing. This yielded pDress_mSci_antiFRET_P2A_linker-noSacI (internal number 5517).

An insert fragment was amplified from the new plasmid by PCR using primers 11-12, containing the last part of the rhamnose promotor, the ribosome binding site, the adapted kozak sequence, the 6xHis-tag, the thrombin recognition site, the mScarlet-I fluorescent protein, the anti-FRET linker and the P2A sequence. A PCR using Phusion polymerase (Thermo Fisher Scientific) was performed on dual expression plasmid pFHL (Addgene plasmid #129628) using primers 13-14 to generate a backbone fragment containing the Tq-Ca-FLITS calcium sensor, the backbone of the pFHL plasmid, the CMV promotor and the first part of the rhamnose promotor. Both insert and backbone fragment were digested with BamHI and EcoRI to create overhanging ends. The fragments were ligated together and *E. coli* was transformed with the DNA. We named this plasmid pFR_mScarletI-antiFRET-P2A-mTurquoise2_T203Y. Correct construction of the new pFR (Franka Ratio) plasmid was verified by control digestion with NheI and BsrGI and sequencing.

G-Ca-FLITS and intermediate variants were put in the ratioplasmid pFR by restriction (MluI and SacI) and ligation of pFR_mScarletI-antiFRET-P2A-mTurquoise2_T203Y and a pFHL plasmid containing the sensor of interest. Correct insertion was verified by sequencing.

As control, also EGFP and green variants of mTq2 and sfmTq2 were put in the pFR plasmid. The FPs mTq2 and sfmTq2 were subjected to directed mutagenesis to create mTq2_T203Y and sfmTq2_T203Y with primers 15-16. Next, PCR fragments were created of the green turquoise FPs and EGFP by using primers 17-18. PCR fragments and a pFR backbone were digested with BamHI and HindIII, followed by ligation. Correct insertion was verified by sequencing.

Green GECIs were put in pFR for comparison. DNA fragments were created by PCR using pGD-CMV-jGCaMP7c (Addgene plasmid #105320), pcDNA3-Cyto-GCaMP3 (Addgene plasmid #64853), pGP-CMV-GCaMP6s (Addgene plasmid #40753) and CMV-G-GECO1.1 (Addgene plasmid #32445) as template and primers 19-20. The fragments and a pFR backbone were digested with BamHI and HindIII, and ligated together. Correct insertion was verified by sequencing.

Plasmids for expression in organelles

The sensor sequence was amplified by PCR to generate cytoplasmic, nuclear, mitochondrial and ER-targeted versions of G-Ca-FLITS for transient transfection of mammalian cells, using primers 21-22 (for ER) or 21 and 23 (all others). DNA fragments and vectors carrying the desired tag were digested with AgeI and BsrGI, followed by ligation. Of the following vectors the FP was exchanged for Tq-Ca-FLITS: 3xnlx-mTurquoise2 (Addgene plasmid #98817) for a nuclear tag, 4xmtx-mScarlet-I (Addgene plasmid #98818), ER-mScarletI (Addgene plasmid #137805) and mVenus-N1 (color variant of Addgene plasmid #54843) for an untagged version. The G-Ca-FLITS-3xnes (3x nuclear export sequence) was made by ligating it into 3xNLS-SGFP2(T65G)-T2A-SGFP2-3xNES (Addgene plasmid # 60492) that was cut with NdeI/BsrGI.

Screening of sensors in bacterial lysate

E. coli bacteria were used for screening of green calcium sensor variants. Bacteria expressing (a library of) sensor variants on pFHL plasmids were grown on SKR-agar plates. Colonies visually showing a (bright) green fluorescence (excitation filter 470/40 nm, emission filter 525/50 nm) under a stereomicroscope (MZFLIII, Leica) were selected. Bacteria were grown overnight in 1.5 mL SKR in a deep well plate (732-2893, VWR) covered with an adhesive seal (AB-0718, Thermo Fisher Scientific) at 37 °C while shaking at 280 rpm. The next day, 1 µL of culture was spotted on a SKR plate for storage, and the remainder of the bacteria were harvested by 15 min centrifugation at 2683 ×g in a swing-out centrifuge (5810 R, Eppendorf). Supernatant was removed and 200 µL of lysis buffer (2% Deoxycholic acid in 50 mM Tris-HCl pH 8.0) was added. During optimization of the lysis protocol, the lysis buffer consisted of 50 mM Tris-HCl pH 8.0 with DOC (0, 1, 2 or 3%), urea (0, 0.5 or 1 M), lysozyme (1 mg/mL), Sodium dodecyl sulfate (0, 0.1, 0.25 or 0.5%) and/or Triton-X100 (0, 0.1, 0.5 or 1%). Bacteria were brought in suspension using a vortex and incubated for 15 minutes at RT while intermittently shaking. If indicated, the plate with suspensions was frozen at

-20 °C and thawed. Cell debris was removed by 40 min centrifugation at 2683 ×g in a swing-out centrifuge. The bacterial lysate was collected and stored overnight at 4 °C. The next day, bacterial lysates were diluted 2.5-4×, in 50mM Tris-HCl pH 8.0 to a final volume of 200 µL in 96-well plates with black walls and a glass bottom (89626, Ibidi).

A JASCO FP-8500 spectrofluorometer with microplate reader adapter was used to measure the excitation and emission spectra. Excitation spectra were measured at 370 - 490 nm with a 2.5 nm bandwidth and 1 nm intervals, while collecting emission at 530 nm with 10 nm bandwidth, using a scan speed of 1000 nm/min, 50 ms response time and a high sensitivity. Emission spectra were measured at 460 - 600 nm with a 2.5 nm bandwidth and 1 nm intervals, while exciting at 425 nm with 5 nm bandwidth, using a scan speed of 1000 nm/min, 0.1 s response time and a high sensitivity. Each spectrum was corrected for background by subtraction of the spectrum of a well with only buffer. The same plate was used for determination of the fluorescence lifetime of the lysates as described under ‘**Lifetime imaging.**’ Spectra and lifetimes were recorded after addition of 0.1 mM CaCl₂ (10 µL of 2 mM) to saturate the sensors, and again in the same wells after addition of 9.5 mM EDTA (10 µL of 0.2 M) to remove all calcium from the sensors. After analysis, interesting colonies were picked and grown for further investigation from the agar plates with spotted colonies.

HeLa cell culture and transfection

HeLa cells (CCL-2) acquired from the American Tissue Culture Collection were maintained in full medium, Dulbecco’s modified Eagle medium + GlutaMAX (61965, Gibco) supplemented with 10% fetal bovine serum (FBS, 10270, Gibco), under 7% humidified CO₂ atmosphere at 37 °C. Cells were washed with Hank’s buffered salt solution (HBSS, 14175, Gibco) and trypsinized (25300, Gibco) for passaging. No antibiotics were used.

HeLa cells were grown in 24-well plates with a glass bottom for imaging (Thermo Fisher Scientific). Transfection mixture was prepared in Opti-MEM (31985047, Thermo Fisher Scientific) with 2 µg Polyethylenimine in water (PEI, pH 7.3, 23966, Polysciences), 50 ng plasmid DNA and 150 ng mock-DNA (empty plasmid), and incubated for 20 min before addition to the cells. Cells were imaged 1- or 2-days post-transfection. Before imaging, the medium was replaced with microscopy medium (137 mM NaCl, 5.4 mM KCl, 1.8 mM CaCl₂, 0.8 mM MgSO₄, 20 mM D-Glucose, 20 mM HEPES pH 7.4) and cells were incubated for 20 min.

BOEC cell culture and microporation

Cord blood Blood Outgrowth Endothelial cells (cbBOEC) were cultivated from healthy donor umbilical cord blood as described before (Martin-Ramirez et al., 2012 [↗](#)). Cells were cultured in Endothelial Cell Growth Medium-2 BulletKit (CC-3162, Lonza) supplemented with 100 U/mL Penicillin (Thermo Fisher Scientific) and 100 µg/mL Streptomycin (Thermo Fisher Scientific) and 20% FBS, under 5% humidified CO₂ atmosphere at 37 °C. cbBOECs were cultured from passage number 4 to 16. Culture dishes and microscopy plates were coated with 0.1% gelatin (Merck) in phosphate-buffered saline 30 min prior to cell seeding.

Cells were transfected by microporation with 2 µg plasmid DNA containing G-Ca-FLITS tagged to various organelles per 3-5 × 10⁵ cells. For microporation, the Neon Transfection System (MPK5000, Invitrogen) and corresponding Neon transfection kit were used according to manufacturers’ protocol. We used the R buffer from the kit, the 100 µL tips, and a 30 ms pulse of 1300 V. After microporation, BOECs were directly seeded in 24-well plates with a glass bottom (Thermo Fisher Scientific). After two hours, the medium was replaced. Cells were imaged 24 h after transfection in full medium.

Frequency domain Lifetime imaging

For FD-FLIM lifetime measurements, we used a Lambert Instruments FLIM Attachment (LIFA) setup, composed of an Eclipse Ti microscope (Nikon) with a Lambert Instruments Multi-LED for excitation, a LI²CAM camera, a LIFA signal generator (all Lambert Instruments) to synchronize the light source and the camera, and was controlled by the LI-FLIM software (version 1.2.13). For GFP measurements, a 446 nm light emitting diode (LED) was used, combined with a 448/20 nm or 472/30 nm excitation filter, a 488 nm dichroic mirror and a 520/35 nm band-pass filter. For CFP measurements, a 446 nm LED was used, combined with a 448/20 nm excitation filter, a 442 nm dichroic mirror and a 482/25 nm band-pass filter (all filters from Semrock). The LED and image intensifier were high-frequency modulated at 40 MHz. Alexa488 or EB was used as a reference to calibrate the instrumentation, with a known mono-exponential lifetime of 4.05 ns (Rusinova et al., 2002 [↗](#); Zheng et al., 2018 [↗](#)) or 0.086 ns (Bastiaens et al., 1992 [↗](#); Boens et al., 2007 [↗](#); van Munster and Gadella, 2004 [↗](#)), respectively.

Fluorescence lifetime of sensors in bacterial lysates was recorded at RT or 37 °C using a ×20 (Plan Apo, NA 0.95 air) objective or a ×40 (Plan Apo, NA 0.95 air) objective, collecting 12 phase images and averaging 3×. When measuring lysates in a 96 well-plate, the LIFA software was controlled by a Matlab script (Bindels et al., 2014 [↗](#)) that automatically moves to the position of each of the wells, adjusts the exposure time based on the intensity to a maximum of 500 ms, collects the lifetime stack and saves the data. Fluorescence lifetime in HeLa cells or BOECs was recorded at 37 °C. If needed, 1 μM histamine or a mix of ionomycin (5 μg/mL, I-6800, LClaboratories) and calcium (5 mM) was added. Cells were imaged using a ×40 (Plan Apo, NA 0.95 air) objective and collecting 12 phase images.

Recorded sample stacks and a reference stack were converted into lifetime images by an ImageJ macro (Bindels et al., 2020 [↗](#); Gadella et al., 1994 [↗](#)). When the intensity of fluorescence in the cells was very weak, a background correction was performed on the lifetime stacks, using a manually indicated background region using Matlab (R2015a). For bacterial lysates, the average phase and modulation lifetime (τ_ϕ and τ_M) of the full view were extracted by the imageJ macro described above. For cells, the lifetime of individual cells was collected. This was done in a semi-automatic manner, using first an ImageJ macro that masks the image based on intensity, followed by a macro that guides the measurements of the cells and the saving of the ROIs. For signal-to-noise calculations, a rectangular ROI was drawn manually inside cells with an area of 400 pixels (20×20, 10×40 or 5×75 pixels), from which the mean and standard deviation were extracted.

Protein isolation

Plasmid were prepared for isolation of the proteins mTq2_T203Y, G-Ca-FLITS and intermediate variant Tq-Ca-FLITS_T203Y. To this end, mScarlet-I, the anti-FRET linker and the P2A site were removed from the pFR plasmids containing the proteins of interest by digestion with NheI followed by ligation. The new plasmids were named pFPO-mTq2_T203Y, pFPO-G-Ca-FLITS and pFPO-Tq-Ca-FLITS_T203Y (FPO being an abbreviation for Franka Protein Only) and correct removal was verified by sequencing.

E. coli bacteria containing a pFPO plasmid were grown overnight in 50 mL SKR medium at 37 °C while shaking. The next morning, the cultures were put at 20 °C for 6 hours, while shaking. Bacteria were harvested by centrifugation for 30 min at 3220 ×g (5810 R, Eppendorf) at 4 °C and washed in 20 mL ice cold ST buffer (20 mM Tris, 200 mM NaCl, pH 8.0), followed by a second centrifugation. The bacteria were resuspended in 5 mL ST buffer and stored at -20 °C. Later, the bacteria were thawed on ice and 5 mg Lysozyme, 1 μl Benzonase (25 U/μl) and 50 μl Halt Protease Inhibitor were added (10320015, Thermo Fisher Scientific). From now on, all steps were performed on ice or at 4 °C. The mixture was incubated for at least 30 min followed by addition of 50 μl of 20% Nonidet-P40. The solution was centrifuged for 30 min at 40062 ×g (Sorval Lynx 6000,

Thermo Fisher Scientific) and loaded on 1 mL (bed volume) freshly prepared Ni-NTA resin, according to manufacturers' instructions (88221, Thermo Fisher Scientific). After 1 hour incubation while gently rotating, the beads were washed 7 times with 14 mL ST buffer, each time collecting the beads by centrifugation for 2 min at 700 ×g. The proteins were eluted once with 0.5 mL ST buffer with 0.6 M imidazole, and a second and third time with 0.5 mL ST buffer with 0.2 M imidazole.

Salt was removed from the isolated proteins by PD10 desalting columns (17-0851-01, Cytiva), according to manufacturers' instructions. Small salts and molecules (imidazole) flow faster through the column than the large proteins. The columns were equilibrated with 10 mM Tris-HCl pH 8.0 before loading, and proteins were eluted with the same buffer. The visually brightest fractions were collected. For storage, aliquots were snap-frozen in liquid nitrogen and stored at -80 °C.

Quantum yield

The quantum yield (QY) was determined by measuring the absorbance and emission spectra of a range of dilutions of purified G-Ca-FLITS or Tq-Ca-FLITS_T203Y in 1 mL 0 or 39 μM free Ca²⁺ buffer of Calcium Calibration Buffer Kit #1 (C3008MP, Thermo Fisher Scientific). In addition, mTq2_T203Y was measured diluted in a 10 mM Tris-HCl buffer (pH 8.0). First, a quartz cuvette was filled with 1 mL of the required buffer and the absorbance spectrum (250 - 550 nm, 1 nm step size) was measured using a spectrophotometer (Libra S70, Biochrom). The cuvette was transferred to a JASCO FP-8500 spectrofluorometer and the emission spectrum was recorded at 460 - 700 nm with a 2.5 nm bandwidth and 1 nm intervals, while exciting at 450 nm with 2.5 nm bandwidth, using a scan speed of 500 nm/min, 1 s response time and a medium sensitivity. A small amount of purified protein was added, and the absorbance and emission spectra were recorded again. This was repeated for in total 4 dilutions per protein and while the absorbance at 450 nm was always below 0.05 for the highest concentration. An additional absorbance spectrum was recorded with a peak absorbance “A_{peak}” of 0.5. The same procedure was followed for rhodamine as a reference in 1 mM NaOH.

Absorbance spectra were corrected by subtraction of the offset of the spectrum between 541 and 550 nm, followed by subtraction of the spectrum of only buffer. The resolution of the spectrophotometer is too low for the low protein levels, so the resulting spectra are not smooth. Therefore, the additionally taken absorbance spectrum with A_{peak} = 0.5 were fitted to the low-resolution spectra. From this fit, the absorbance at 450 nm (A₄₅₀) was determined for each dilution of protein. Emission spectra were corrected for spectral sensitivity of the detector and the spectrum of the only buffer sample was subtracted. The spectral area “I_{em}” under corrected emission spectra was calculated by integration between 460 - 700 nm.

The “A₄₅₀” was plotted versus “I_{em}” and the slope “s” was determined while forcing the regression line through the origin: A₄₅₀ = s × I_{em}. The regression was performed in R studio (version 1.0.136), using the linear model from the stats package with default parameters. The Quantum Yield (QY_s) of the protein samples was determined according to:

$$QY_s = QY_r \times \frac{s_s}{s_r} \quad (1)$$

where subscripts “s” and “r” indicate the protein sample and the reference rhodamine, QY_r = 0.85 (Zhang et al., 2014 [DOI](#)), respectively. The standard deviation in “QY_s” was determined from the standard deviation of “s_s”.

Extinction coefficient

Purified G-Ca-FLITS, Tq-Ca-FLITS_T203Y and mTq2_T203Y were $\sim 10\times$ diluted in calcium buffers containing 0 or 39 μM free calcium of the Calcium Calibration Buffer Kit #1 (C3008MP, Thermo Fisher Scientific) to a final volume of 200 μL . The absorbance spectra were measured before and after addition of 0.5 M NaOH and 2M urea, at 260 - 650 nm with 1 nm step size and 1 nm bandwidth. Corresponding buffer was used as reference. The concentration of unfolded protein was determined using the Beer-Lambert law and assuming an extinction coefficient (ϵ) at 462 nm of $46 \text{ mM}^{-1} \text{ cm}^{-1}$ for the free cyan chromophore (Lelimosin et al., 2009). Next, ϵ_{max} was determined at maximum absorbance for G-Ca-FLITS, Tq-Ca-FLITS_T203Y and mTq2_T203Y.

In vitro calibration

Purified G-Ca-FLITS were diluted $1000\times$ in calcium buffers ranging from 0 to 39 μM , using the Calcium Calibration Buffer Kit #1 according to manufacturers' instructions. Dilutions were made in triplicate in a 96-well plate with black walls and flat glass bottom (89626, Ibidi). Fluorescence lifetime was recorded of each well as described under 'lifetime imaging', first at RT and later at 37 °C, using the same plate. The average phase and modulation lifetime (τ_{ϕ} and τ_M) of the full view were determined per well. The average τ_{ϕ} and τ_M were determined for 0 and 39 μM free Ca^{2+} , these are "min" and "max". The Phase (Φ) and modulation (M) were calculated from the recorded lifetimes and displayed in a polar plot as G and S coordinates:

$$\begin{aligned}\Phi &= \text{atan}(\omega\tau_{\phi}) & \text{and} & & M &= \sqrt{\frac{1}{1+(\omega\tau_M)^2}} \\ G &= M \cos(\Phi) & \text{and} & & S &= M \sin(\Phi)\end{aligned}\quad (2)$$

The angular frequency of modulation, $2\pi f$, is given by " ω ", and we used $f = 40 \text{ MHz}$. Measurements were projected on the straight line between the two extremes ("min" and "max") and converted to line fraction " a " (equation 3).

$$\begin{aligned}dG &= G - G_{\min} & \text{and} & & dS &= S - S_{\min} \\ a &= \frac{dG \times dG_{\max} + dS \times dS_{\max}}{dG_{\max}^2 + dS_{\max}^2}\end{aligned}\quad (3)$$

The line fraction was corrected of for the intensity contribution of the two states to find the true fraction " F ", with $F = 1$ representing all sensors in the calcium-bound state (Eq. 4). The intensity ratio (R) between states in vitro was determined to be $R = 0.836$, based on the theoretical ratio calculated from the spectra of the sensor and the instrument settings, see Supplemental Note 1).

$$F = \frac{a}{R \times (1-a) + a}\quad (4)$$

The fraction " F " was fitted with the Hill equation (equation 5) to find the in vitro K_d , using the Nonlinear Least Squares method of the R Stats Package (version 3.3.3) in R Studio (version 1.0.136), using the port algorithm.

$$F = F_{\min} + \frac{(F_{\max} - F_{\min})}{\left(\frac{K_d}{L}\right)^n + 1}\quad (5)$$

The microscopic dissociation constant is given by " K_d ", the known free Ca^{2+} concentration by " L " and the Hill coefficient by " n ".

The 95% confidence interval (CI) of the fraction of the two extremes was determined from the data. The mean of the low calcium state plus the CI was determined to be the lowest measurable fraction, the mean of the high calcium state minus the CI was determined to be the highest

measurable fraction. From this, the lowest and highest measurable concentration was determined using the Hill equation (equation 5 [↗](#)).

pH sensitivity

A series of buffers ranging from pH 2.8 to 10.1 were prepared, using 50 mM citrate buffer (pH 2.8 - 5.8), MOPS buffer (pH 6.3 - 7.9) and glycine/NaOH buffer (pH 8.2 - 10.1). Buffers additionally contain 0.1 M KCl and 0.1 mM CaCl_2 or 5 mM EGTA. The pH of each buffer was determined including all components shortly before use. Purified G-Ca-FLITS or mTq2_T203Y was diluted 520× in the prepared buffers, using a predilution of 40× and a total volume of 200 μL . The dilutions were made in triplicate in a 96-well plate with black walls and flat glass bottom (89626, ibidi). The lifetime of each well in the same 96-well plate was recorded at RT, see ‘Lifetime imaging.’

Ratiometric imaging

HeLa cells co-expressing a green calcium sensor (or green FP) and mScarlet-I form a single plasmid (pFR) were imaged 24 h after transfection at 37 °C, using an Eclipse Ti microscope (Nikon) equipped with a Spectra X Light Engine (Lumencor) for excitation and an Orca flash 4.0 camera (Hamamatsu). Cells were imaged using a Plan Apo ×10 NA 0.45 air objective. For imaging of GFP, a 470 nm LED, a 470/24 nm excitation filter and a quad band cube (MXU 71640, Nikon) were used. For RFP, a 575 nm LED, a 575/25 nm excitation filter and the same quad band cube were used. Only the middle of the field of view was used, to minimize the unequal illumination. A 4×4 tile scan was performed and channels were imaged sequentially.

Fluorescence was recorded without stimulation and directly after stimulation with a mix of ionomycin (5 $\mu\text{g/mL}$, I-6800, LClaboratories) and calcium (5 mM). Background was subtracted from the images and the average green and red fluorescence intensity of individual cells was measured using ImageJ (version 1.52k). The ratio GFP/RFP was calculated for each cell. As a control, we used a sample with only mScarlet-I (ratio GFP/RFP = 0) and a sample with EGFP and mScarlet-I (ratio GFP/RFP = 1). The ratio of all cells was normalized to the controls. The experiment was performed twice per construct.

Confocal imaging

Confocal images of HeLa cells and BOECs expressing G-Ca-FLITS were taken with a Leica Sp8 system (Leica) with a ×63 objective (HL PL Apo, C2S NA 1.40 oil) at 37 °C and 7% (HeLa cells) or 5% CO_2 (BOECs). HeLa cells were imaged in MM and BOECs in growth medium, as stated before. A 488 nm Argon laser was used for imaging, combined with an Acousto-Optical Beam Splitter (AOBS), and fluorescence was collected with a HyD at 500 - 780 nm. The pinhole was set to 1 AU.

Confocal TCSPC imaging - PicoHarp TCSPC module

Confocal fluorescence lifetime images of G-Ca-FLITS expressed in mitochondria of HeLa cells were acquired at an FV1000 (Olympus) confocal microscope equipped with a PicoHarp TCSPC module (PicoQuant). A field of view of 256×256 pixels (207 - 331 nm per pixel) was illuminated with a pulsed 485 nm Picoquant diode laser (20 MHz, 0.4 $\text{kW}\cdot\text{cm}^{-2}$) using an Olympus UPLS Apo 60x water NA1.2 objective lens. The fluorescence signal was detected in confocal mode with the pinhole diameter set at 130 μm . The fluorescence passed a 405/480/560/635 dichroic mirror, was filtered by a 505 - 540 nm emission filter and detected by avalanche photodiodes (MPD).

In order, to obtain a reliable fluorescence lifetime the measurement times of the sample channel was set such that at least 10^5 photons were collected. An image of the median arrival times per pixel was inspected for heterogeneity. Based on the image, one or more regions were manually selected for fitting of a decay curve. The full fluorescence decay curves (5-6 minutes integration time) were fitted in the Symphotime64 software (PicoQuant) using a bi-exponential decay model including an instrumental response function generated from the same dataset. On basis of visual

inspection of the fit, the fit residuals and the minimal Chi square the fitted results were accepted or discarded. The fitting of the decay curves was repeated for all areas, but now an integration time of 1 minute was used. The fitting was repeated for every minute, to obtain information about temporal variation.

Confocal TCSPC imaging – Leica Stellaris

Confocal fluorescence lifetime images of G-Ca-FLITS expressed in mitochondria of HeLa cells were acquired with a Leica Stellaris8 equipped with a pulsed white light laser set to 40 MHz and a 40x NA 0.95 air PL APO objective. The excitation was at 474 nm and the emission was detected between 490 nm and 600 nm by the HyDX2 detector. The pinhole was set to 4.4 airy units, the imaging speed to 400 Hz per line (pixel dwell time 3.16 μ s) with 8 line repetitions and a frame size of 512×512 (pixel size 0.259 μ m). For live-cell imaging, the images were acquired every minute.

The data was saved as .ptu file and a custom script was used to convert these data into G and S images. The G and S data, together with the K_D determined by the calibration, were used to convert the data into calcium concentration.

Imaging and analysis of 2P-FLIM of *Drosophila*

We imaged the brain of *Drosophila melanogaster* females attached to a custom physiology platform (Maimon et al, 2010 [↗](#), Mussells Pires et al., 2024 [↗](#)). This platform keeps the fly's body dry and in air, with the brain exposed to an experimentally determined saline. The flies were freely dangling and we did not track their behavior. We removed the superior cuticle of the head (the frons and anterior ocellar cuticle) to access the brain dorsally for optical imaging. Extracellular saline was prepared as reported (Mussells Pires et al., 2024 [↗](#)), consisting of (in mM): 103 NaCl, 3 KCl, 5 TES, 10 trehalose dihydrate, 10 glucose, 2 sucrose, 26 NaHCO₃, 1 NaH₂PO₄, 1.5 CaCl₂ and 4 MgCl₂ titrated with Milli-Q water to an osmolarity of 280-285 mOsm. The high [K⁺] solution exchanged the concentrations of NaCl with KCl, keeping the osmolarity and [Cl⁻] unchanged.

The saline was kept at ~21°C for the duration of all experiments using a Warner Instruments temperature controller (CL-100) driving a Peltier device (SC-20). The Peltier device was placed in-line with tubing (Tygon 2375) that perfused the preparation with saline at a rate of ~3 mL per minute. A thermistor in the bath was used to regulate the Peltier device in closed loop. The high [K⁺] solution was applied by allowing flow through the perfusion line to exchange with the extracellular solution, which was gradually aspirated out of the chamber using a glass pipette attached to a vacuum. Each recording was given ~10 minutes to respond to the new solution and we estimate that the saline composition of the preparation was fully replaced within ~1 min.

The 2P-FLIM setup uses an Olympus 20x 1.0 NA water immersion objective (XLUMPLFLN) on a customized Sutter MIMMs to be detailed in a later manuscript by SCT and GM. A Coherent Discovery NX laser's tunable beam set to 920 nm (with -12,500 fs² of dispersion compensation) was used to excite all three fluorophores.

Frames were collected at a resolution of 256 × 256 pixels and the time series data shown in **Figure 8** [↗](#) are plotted at ~1 Hz. Flies expressing jGCaMP7f or Tq-Ca-FLITS were imaged with ~20 mW of laser power measured after the objective, while flies expressing G-Ca-FLITS were imaged with ~30 mW laser power.

The fluorescence lifetime was computed by fitting the first 1200 frames to a biexponential distribution convolved with a Gaussian plus randomly distributed uniform noise by minimizing the chi-squared statistic (Thornquist et al., 2020). The offset of this fit was used to adjust the mean photon arrival time within each frame (empirical lifetime), and the contribution of the estimated noise was subtracted out. Explicitly, the arrival time histogram was fit to the equation:

$$p(t) = \epsilon + f_1 F_1(t; \tau_1, \sigma, \mu) + f_2 F_2(t; \tau_2, \sigma, \mu) \quad (6)$$

where

$$F(t; \tau, \sigma, \mu) = \sqrt{\frac{\pi \sigma^2}{2 \tau^2}} \exp\left(-\frac{\sigma^2}{2 \tau^2}\right) \operatorname{erfc}\left(\frac{1}{\sqrt{2}}\left(\frac{\sigma}{\tau} - \frac{t - \mu}{\sigma}\right)\right) \exp\left(-\frac{t - \mu}{\tau}\right) \quad (7)$$

is the single exponential response with a time constant τ convolved with a Gaussian centered at μ with standard deviation σ and $\epsilon + f_1 + f_2 = 1$. To compute the empirical lifetime L of a collection of photons we computed the mean arrival time of the photons $\langle p \rangle = \sum_{t=0}^{t_{\max}} t * p(t)$, then removed the offset and the estimated noise due to the uniform signal (which had mean arrival time $\frac{t_{\max}}{2}$ because it is uniformly distributed):

$$L = \langle p \rangle - \left(\mu + \epsilon \frac{t_{\max}}{2}\right) \quad (8)$$

Mean intensity-weighted lifetime refers to the following procedure: when taking the mean of a set of n frames, the lifetime was computed as

$$\langle L \rangle = \frac{\sum_{k=1}^n I_k L_k}{n \langle I \rangle} \quad (9)$$

where I_k refers to the intensity of the k th frame, L_k refers to the mean lifetime of the k th frame, and $\langle \circ \rangle$ refers to the average across frames.

Data availability

Data is available at zenodo, DOI: 10.5281/zenodo.17237551 <https://doi.org/10.5281/zenodo.17237551>

Several plasmids are deposited for distribution through Addgene (www.addgene.org). The plasmids and corresponding addgene numbers are: N1-G-Ca-FLITS: #191465, 3xnl-G-Ca-FLITS: #191463, 4xmt-G-Ca-FLITS: #191462, ER-G-Ca-FLITS-KDEL: #191464, pFPO-His-mTurquoise2_T203Y: #191456, pFPO-His-G-Ca-FLITS: #191455, pFR-mScarletI-antiFRET-P2A-G-Ca-FLITS: #191457, pFR-mScarletI-antiFRET-P2A-EGFP: #191458, pFR-mScarletI-antiFRET-P2A-mTurquoise2_T203Y: #191459, pFR-mScarletI-antiFRET-P2A-GCaMP3: #191460, pFR-mScarletI-antiFRET-P2A-jGCaMP7c: #191461, pDress-mScarletI-antiFRET-P2A2-linker_noSacI: #191473. Transgenic *Drosophila* lines expressing Tq-Ca-FLITS under UAS or LexA control and G-Ca-FLITS under UAS control are available upon request.

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Additional information

Author contributions

F.H.L. and J.G. conceptualized the project, designed the experiments, interpreted the results, and wrote the manuscript. F.H.L. engineered and characterized the green biosensor, and tested it in mammalian cells. S.C.T. and G.M. performed experiments on *Drosophila*, analysed the data and prepared text and figures. R.M.B. and J.Y.H. were involved in the engineering of the green biosensor and setting up the screen. M.H. assisted with TCSPC FLIM imaging and processing of the data. T.W.J.G. assisted in FLIM data processing, analysis, and interpretation of the data. All authors approved the final manuscript.

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Additional files

Supplemental Material 

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Author information

Franka H van der Linden

Swammerdam Institute for Life Sciences, Section of Molecular Cytology, van Leeuwenhoek Centre for Advanced Microscopy, University of Amsterdam, Amsterdam, Netherlands
ORCID iD: [0000-0001-5324-2714](https://orcid.org/0000-0001-5324-2714)

Stephen C Thornquist

Laboratory of Integrative Brain Function and Howard Hughes Medical Institute, The Rockefeller University, New York, United States
ORCID iD: [0000-0003-2693-4971](https://orcid.org/0000-0003-2693-4971)

Rick M ter Beek

Swammerdam Institute for Life Sciences, Section of Molecular Cytology, van Leeuwenhoek Centre for Advanced Microscopy, University of Amsterdam, Amsterdam, Netherlands

Jelle Y Huijts

Swammerdam Institute for Life Sciences, Section of Molecular Cytology, van Leeuwenhoek Centre for Advanced Microscopy, University of Amsterdam, Amsterdam, Netherlands

Mark A Hink

Swammerdam Institute for Life Sciences, Section of Molecular Cytology, van Leeuwenhoek Centre for Advanced Microscopy, University of Amsterdam, Amsterdam, Netherlands

Theodorus WJ Gadella Jr.

Swammerdam Institute for Life Sciences, Section of Molecular Cytology, van Leeuwenhoek Centre for Advanced Microscopy, University of Amsterdam, Amsterdam, Netherlands
ORCID iD: [0000-0002-7639-219X](https://orcid.org/0000-0002-7639-219X)

Gaby Maimon

Laboratory of Integrative Brain Function and Howard Hughes Medical Institute, The Rockefeller University, New York, United States
ORCID iD: [0000-0003-1219-5856](https://orcid.org/0000-0003-1219-5856)

Joachim Goedhart

Swammerdam Institute for Life Sciences, Section of Molecular Cytology, van Leeuwenhoek Centre for Advanced Microscopy, University of Amsterdam, Amsterdam, Netherlands
ORCID iD: [0000-0002-0630-3825](https://orcid.org/0000-0002-0630-3825)

For correspondence: j.goedhart@uva.nl

Editors

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Reviewer #1 (Public review):

Summary:

van der Linden et al. report on the development of a new green-fluorescent sensor for calcium, following a novel rational design strategy based on the modification of the cyan-emissive sensor mTq2-CaFLITS. Through a mutational strategy similar to the one used to convert EGFP into EYFP, coupled with optimization of strategic amino acids located in proximity of the chromophore, they identify a novel sensor, G-CaFLITS. Through a careful characterization of the photophysical properties in vitro and the expression level in cell cultures, the authors demonstrate that G-CaFLITS combines a large lifetime response with a good brightness in both the bound and unbound states. This relative independence of the brightness on calcium binding, compared with existing sensors that often feature at least one very dim form, is an interesting feature of this new type of sensors, which allows for a more robust usage in fluorescence lifetime imaging. Furthermore, the authors evaluate the performance of G-CaFLITS in different subcellular compartments and under two-photon excitation in *Drosophila*. While the data appears robust and the characterization thorough, the interpretation of the results in some cases appears less solid, and alternative explanations cannot be excluded.

Strengths:

The approach is innovative and extends the excellent photophysical properties of the mTq2-based to more red-shifted variants. While the spectral shift might appear relatively minor, as the authors correctly point out, it has interesting practical implications, such as the possibility to perform FLIM imaging of calcium using widely available laser wavelengths, or to reduce background autofluorescence, which can be a significant problem in FLIM.

The screening was simple and rationally guided, demonstrating that, at least for this class of sensors, a careful choice of screening positions is an excellent strategy to obtain variants with large FLIM responses without the need of high-throughput screening.

The description of the methodologies is very complete and accurate, greatly facilitating the reproduction of the results by others, or the adoption of similar methods. This is particularly true for the description of the experimental conditions for optimal screening of sensor variants in lysed bacterial cultures.

The photophysical characterization is very thorough and complete, and the vast amount of data reported in the supporting information is a valuable reference for other researchers willing to attempt a similar sensor development strategy. Particularly well done is the characterization of the brightness in cells, and the comparison on multiple parameters with existing sensors.

Overall, G-CaFLITS displays excellent properties for a FLIM sensor: very large lifetime change, bright emission in both forms and independence from pH in the physiological range.

Comment on revised version:

The authors have significantly improved the manuscript, and overall I fully agree in maintaining the assessment as it is now.

<https://doi.org/10.7554/eLife.105086.2.sa3>

Reviewer #2 (Public review):

Summary:

Van der Linden et al. describe the addition of the T203Y mutation to their previously described fluorescence lifetime calcium sensor Tq-Ca-FLITS to shift the fluorescence to green emission. This mutation was previously described to similarly red-shift the emission of green and cyan FPs. Tq-Ca-FLITS_T203Y behaves as a green calcium sensor with opposite polarity compared with the original (lifetime goes down upon calcium binding instead of up). They then screen a library of variants at two linker positions and identify a variant with slightly improved lifetime contrast (Tq-Ca-FLITS_T203Y_V27A_N271D, named G-Ca-FLITS). The authors then characterize the performance of G-Ca-FLITS relative to Tq-Ca-FLITS in purified protein samples, in cultured cells, and in the brains of fruit flies.

Strengths:

This work is interesting as it extends their prior work generating a calcium indicator scaffold for fluorescent protein-based lifetime sensors with large contrast at a single wavelength, which is already being adopted by the community for production of other FLIM biosensors. This work effectively extends that from cyan to green fluorescence. While the cyan and green sensors are not spectrally distinct enough (~20-30nm shift) to easily multiplex together, it at least shifts the spectra to wavelengths that are more commonly available on commercial microscopes.

The observations of organellar calcium concentrations were interesting and could potentially lead to new biological insight if followed up.

<https://doi.org/10.7554/eLife.105086.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors present a variant of a previously described fluorescence lifetime sensor for calcium. Much of the manuscript describes the process of developing appropriate assays for screening sensor variants, and thorough characterization of those variants (inherent fluorescence characteristics, response to calcium and pH, comparisons to other calcium sensors). The final two figures show how the sensor performs in cultured cells and in vivo drosophila brains.

Strengths:

The work is presented clearly and the conclusion (this is a new calcium sensor that could be useful in some circumstances) is supported by the data.

Weaknesses:

There are probably few circumstances where this sensor would facilitate experiments (calcium measurements) that other sensors would prove insufficient.

Comment on revised version:

I think the manuscript has been significantly improved and I concur with the eLife Assessment statement.

[Editors' note: There are no further requests by the reviewers. All of them expressed their approval of the new version of the manuscript.]

<https://doi.org/10.7554/eLife.105086.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

van der Linden et al. report on the development of a new green-fluorescent sensor for calcium, following a novel rational design strategy based on the modification of the cyan-emissive sensor mTq2-CaFLITS. Through a mutational strategy similar to the one used to convert EGFP into EYFP, coupled with optimization of strategic amino acids located in proximity of the chromophore, they identify a novel sensor, G-CaFLITS. Through a careful characterization of the photophysical properties in vitro and the expression level in cell cultures, the authors demonstrate that G-CaFLITS combines a large lifetime response with a good brightness in both the bound and unbound states. This relative independence of the brightness on calcium binding, compared with existing sensors that often feature at least one very dim form, is an interesting feature of this new type of sensors, which allows for a more robust usage in fluorescence lifetime imaging.

Furthermore, the authors evaluate the performance of G-CaFLITS in different subcellular compartments and under two-photon excitation in *Drosophila*. While the data appears robust and the characterization thorough, the interpretation of the results in some cases appears less solid, and alternative explanations cannot be excluded.

Strengths:

The approach is innovative and extends the excellent photophysical properties of the mTq2-based to more red-shifted variants. While the spectral shift might appear relatively minor, as the authors correctly point out, it has interesting practical implications, such as the possibility to perform FLIM imaging of calcium using widely available laser wavelengths, or to reduce background autofluorescence, which can be a significant problem in FLIM.

The screening was simple and rationally guided, demonstrating that, at least for this class of sensors, a careful choice of screening positions is an excellent strategy to obtain variants with large FLIM responses without the need of high-throughput screening.

The description of the methodologies is very complete and accurate, greatly facilitating the reproduction of the results by others, or the adoption of similar methods. This is particularly true for the description of the experimental conditions for optimal screening of sensor variants in lysed bacterial cultures.

The photophysical characterization is very thorough and complete, and the vast amount of data reported in the supporting information is a valuable reference for other researchers willing to attempt a similar sensor development strategy. Particularly well done is the characterization of the brightness in cells, and the comparison on multiple parameters with existing sensors.

Overall, G-CaFLITS displays excellent properties for a FLIM sensor: very large lifetime change, bright emission in both forms and independence from pH in the physiological range.

Weaknesses:

The paper demonstrates the application of G-CaFLITS in various cellular subcompartments without providing direct evidence that the sensor's response is not affected by the targeting. Showing at least that the lifetime values in the saturated state are similar in all compartments would improve the robustness of the claims.

*In some cases, the interpretation of the results is not fully convincing, leaving alternative hypotheses as a possibility. This is particularly the case for the claim of the origin of the strongly reduced brightness of G-CaFLITS in *Drosophila*. The explanation of the intensity changes of G-CaFLITS also shows some inconsistency with the basic photophysical characterization.*

While the claims generally appear robust, in some cases they are conveyed with a lack of precision. Several sentences in the introduction and discussion could be improved in this regard. Furthermore, the use of the signal-to-noise ratio as a means of comparison between sensors appears to be imprecise, since it is dependent on experimental conditions.

We thank the reviewer for a thorough evaluation and for suggestions to improve our manuscript. We are happy with the recognition of the strengths of this work. The list with weaknesses has several valid points which will be addressed in a point-by-point reply and a revision.

Reviewer #2 (Public review):

Summary:

Van der Linden et al. describe the addition of the T203Y mutation to their previously described fluorescence lifetime calcium sensor Tq-Ca-FLITS to shift the fluorescence to green emission. This mutation was previously described to similarly red-shift the emission of green and cyan FPs. Tq-Ca-FLITS_T203Y behaves as a green calcium sensor with opposite polarity compared with the original (lifetime goes down upon calcium binding instead of up). They then screen a library of variants at

two linker positions and identify a variant with slightly improved lifetime contrast (TqCa-FLITS_T203Y_V27A_N271D, named G-Ca-FLITS). The authors then characterize the performance of G-Ca-FLITS relative to Tq-Ca-FLITS in purified protein samples, in cultured cells, and in the brains of fruit flies.

Strengths:

This work is interesting as it extends their prior work generating a calcium indicator scaffold for fluorescent protein-based lifetime sensors with large contrast at a single wavelength, which is already being adopted by the community for production of other FLIM biosensors. This work effectively extends that from cyan to green fluorescence. While the cyan and green sensors are not spectrally distinct enough (~20-30nm shift) to easily multiplex together, it at least shifts the spectra to wavelengths that are more commonly available on commercial microscopes.

The observations of organellar calcium concentrations were interesting and could potentially lead to new biological insight if followed up.

Weaknesses:

(1) The new G-Ca-FLITS sensor doesn't appear to be significantly improved in performance over the original Tq-Ca-FLITS, no specific benefits are demonstrated.

(2) Although it was admirable to attempt in vivo demonstration in Drosophila with these sensors, depolarizing the whole brain with high potassium is not a terribly interesting or physiological stimulus and doesn't really highlight any advantages of their sensors; G-Ca-FLITS appears to be quite dim in the flies.

We thank the reviewer for a thorough evaluation and for suggestions to improve our manuscript. Although the spectral shift of the green variant is modest, we have added new data (figure 7) to the manuscript that demonstrates multiplex imaging of G-Ca-FLITS and Tq-Ca-FLITS.

As for the listed weaknesses we respond here:

(1) Although we agree that the performance in terms of dynamic range is not improved, the advantage of the green sensor over the cyan version is that the brightness is high in both states.

(2) We agree that the performance of G-Ca-FLITS is disappointing in Drosophila. We feel that this is important data to report, and it makes it clear that Tq-Ca-FLITS is a better choice for this system. Depolarization of the entire brain was done to measure the maximal lifetime contrast.

Reviewer #3 (Public review):

Summary:

The authours present a variant of a previously described fluorescence lifetime sensor for calcium. Much of the manuscript describes the process of developing appropriate assays for screening sensor variants, and thorough characterization of those variants (inherent fluorescence characteristics, response to calcium and pH, comparisons to other calcium sensors). The final two figures show how the sensor performs in cultured cells and in vivo drosophila brains.

Strengths:

The work is presented clearly and the conclusion (this is a new calcium sensor that could be useful in some circumstances) is supported by the data.

Weaknesses:

There are probably few circumstances where this sensor would facilitate experiments (calcium measurements) that other sensors would prove insufficient.

We thank the reviewer for the evaluation of our manuscript. As for the indicated weakness, we agree that the main application of genetically encoded calcium biosensors is to measure qualitative changes in calcium. However, it can be argued that due to a lack of tools the absolute quantification has been very challenging. Now, thanks to large contrast lifetime biosensors the quantitative measurements are simplified, there are new opportunities, and the probe reported here is an improvement over existing probes as it remains bright in both states, further improving quantitative calcium measurements.

Reviewer #1 (Recommendations for the authors):

While the science in the paper appears solid, the methods well grounded and excellently documented, the manuscript would benefit from a revision to improve the clarity of the exposition. In particular:

Part of the introduction appears like a patchwork of information with poor logical consequentiality. The authors rapidly pass from the impact of brightness on FLIM accuracy, to mitochondrial calcium in pathology, to the importance of the sensor's affinity, to a sentence on sensor's kinetics, to fluorescent dyes and bioluminescence, to conclude that sensors should be stable at mitochondrial pH. I highly recommend rewriting this part.

We thank the referee for the comment and we have adjusted to introduction to better connect the parts and increase the logic. The updated introduction addresses all the feedback by the reviewers on different aspects of the introductory text, and we have removed the section on dyes and bioluminescence. We feel that the introduction is better structured now.

The reference to particular amino acid positions would greatly benefit from including images of the protein structure in which the positions are highlighted, similar to what the same authors do in their fluorescent protein development papers. While in the case of sensors a crystal structure might be lacking, highlighting the positions with respect to an AlphaFold-generated structure or the structure of mTq2 might still be helpful.

We appreciate this remark and we have added a sequence alignment of the FLITS probes to supplemental Figure S4. This shows the residues with number, and we have also highlighted the different domains, linkers and mutations. We think that this linear representation works

better than a 3D structure (one issue is that alphafold fails to display the chromophore and it has usually poor confidence for linker residues).

The use of SNR, as defined by the authors (mean of the lifetime divided by standard deviation) appears a poorly suited parameter to compare sensors, as it depends on the total number of collected photons and on the strength of the algorithms used to retrieve the lifetime value. In an extreme example, if one would collect uniform images with millions of photons per pixel, most likely SNR would be extremely good for all sensors in all states, irrespective of the fact that some states are dimmer (within reasonable limits). On the other hand, if the same comparison would be performed at a level of thousands or hundreds of photons per pixel, the effect of different brightness on the SNR would be much more dramatic. While in general I fully agree with the core concept of the paper, i.e. that avoiding low-brightness forms leads more easily to experiments with higher SNR, I would suggest to stick to comparing the sensors in terms of brightness and refer to SNR (if needed) only when describing the consequences on measurements.

The reviewer is right that in absolute terms the SNR is not meaningful. In addition to acquisition time, it depends on expression levels. Yet, it is possible to compare the change in SNR between the apo- and saturated states, and that is what is shown in figure 5. We have added text to better explain that the change in SNR is relevant here:

“The absolute SNR is not relevant here, as it will depend on the expression level and acquisition time. But since we have measured the two extremes in the same cells, we can evaluate how the SNR changes between these states for each separate probe”

Some statements from the authors or aspects of the paper appear problematic:

(1) "Additionally, the fluorescence of most sensors is a non-linear function of calcium concentration, usually with Hill coefficients between 2 and 3. This is ideal when the probe is used as a binary detector for increases in Ca²⁺ concentrations, but it makes robust quantification of low, or even intermediate, calcium concentrations extremely challenging."

To the best of my knowledge, for all sensors the fluorescence response is a nonlinear function of calcium concentrations. If the authors have specific examples in mind in which this is not true, they should cite them specifically. Furthermore, the Hill coefficient defines the range of concentrations in which the sensor operates, while the fact that "low concentrations" might be hard to detect depends only on the dim fluorescence of some sensors in the unbound form.

We agree with the reviewer that this part is not clearly written and confusing, as the sentence “Additionally, the fluorescence of most sensors is a non-linear function of calcium concentration, usually with Hill coefficients between 2 and 3” was not relevant in this section and so we removed it. Now it reads:

“Many GECIs harboring a single fluorescent protein (FP), like GCaMPs, are optimized for a large intensity change, and have a (very) dim state when calcium levels are below the KD of the probe (Akerboom et al., 2013; Dana et al., 2019; Shen et al., 2018; Zhang et al., 2023; Zhao et al., 2011). This is ideal when the probe is used as a binary detector for increases in Ca²⁺ concentrations, but it makes robust quantification of low, or even intermediate, calcium concentrations extremely challenging”

(2) "The affinity of a sensor is of major importance: a low KD can underestimate high concentrations and vice versa."

It is not clear to me why the concentrations would be underestimated, rather than just being less precise. Also, if a calibration curve is plotted in linear scale rather than logarithmic scale, it appears that the precision problem is much more severe near saturation (where low lifetime changes result in large concentration changes) than near zero (where low concentration changes produce large lifetime changes).

We agree that this could be better explained, what we meant to say that concentrations that are ~10x lower or higher than the K_D cannot be precisely measured. See also our reply to the next comment.

(3) "Differences can also arise due to the method of calibration, i.e. when the absolute minimum and maximum signal are not reached in the calibration procedure (Fernandez-Sanz et al., 2019)."

Unless better explained, this appears obvious and not worth mentioning.

What may be obvious to the reviewer (and to us) may not be obvious to the reader, and that's why this is included. To make it clearer we rephrased this part as a list of four items:

"Accurate determination of the affinity of a sensor is important and there are several issues that need to be considered during the calibration and the measurements: (i) the concentrations can only be measured with sufficient precision when it is in the range between $10 \times K_D$ and $1/10 \times K_D$, (ii) the calibration is only valid when the two extremes are reached during the calibration procedure (Fernandez-Sanz et al., 2019), (iii) the sensor's kinetics should be sufficiently fast enough to be able to track the calcium changes, and (iv) the biosensor should be compatible with the high mitochondrial pH of 8 (Cano Abad et al., 2004; Llopis et al., 1998)."

(4) In the experiments depicted in Figure 6C the underlying assumption is that the sensor behaves in the same way independently of the compartment to which it is targeted. This is not necessarily the case. It would be valuable to see the plots of Figure 6C and D discussed in terms of lifetime. Is the saturating lifetime value the same in all compartments?

This is a valid point and we have now included a plot with the actual lifetime data for each of the organelles (figure S15).

We have also added text to discuss this point: "We note that the underlying assumption of the quantification of organellar calcium concentrations is that the lifetime contrast is the same. This is broadly true for most of the measurements (Figure S15). Yet, there are also differences. It is currently unclear whether the discrepancies are due to differences in the physicochemical properties of the compartments, or whether there is a technical reason (the efficiency of ionomycin for saturating the biosensor in the different compartments is unknown, as far as we know). This is something that is worth revisiting. A related issue that deserves attention is the level of agreement between in vitro and in vivo calibrations."

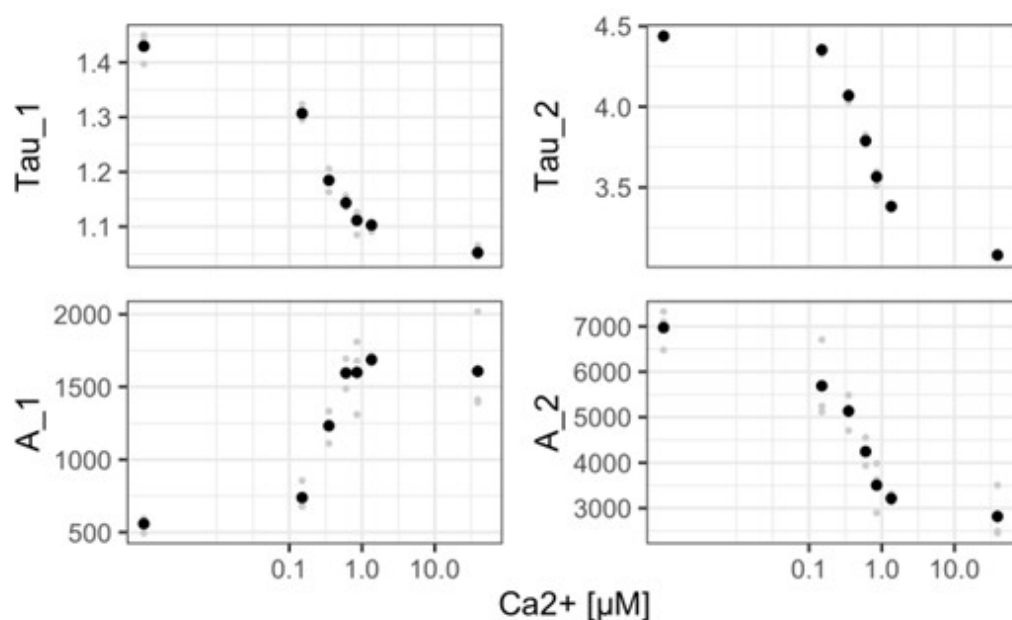
(5) A similar problem arises for the observation of different calcium levels in peripheral mitochondria. In figure S11b, the values of the two lifetime components of a biexponential fit are displayed. Both the long and short components seem to be different. This is an interesting observation, as in an ideal sensor (in which the "long lifetime conformation" is the same whether the sensor is bound to the analyte or not, and similarly for the short lifetime one) those values should be identical. While it is entirely possible that this is not the case for G-CaFLITS, since the authors have conducted a calibration experiment using time-domain FLIM, could they show the behavior of the

lifetimes and preamplitudes? Are the trends consistent with their interpretation of a different calcium level in the two mitochondrial populations?

We have analyzed the calibration data from TCSPC experiments done with the Leica Stellaris. From these data (acquired at high photon counts as it is purified protein in solution), we infer that both the short and long lifetime do change as a function of calcium concentration. In particular the long lifetime shows a substantial change, which we cannot explain at this moment. We agree that this is interesting and may potentially give insight in the conformation changes that give rise to the lifetime change.

The lifetime data of the mitochondria has been acquired with a different FLIM setup, but the trend is consistent, both the long and short lifetime decrease in the peripheral mitochondria that have a higher calcium concentration.

Author response image 1



(6) "The lifetime response of Tq-Ca-FLITS and the $\Delta F/F$ response of jGCaMP7f resembled each other, with both signals gradually increasing over the span of 3-4 minutes after we increased external $[\text{K}^+]$; the two signals then hit a plateau for ~ 1 min, followed by a return to baseline and often additional plateaus (Figure 8B-C). By comparison, G-Ca-FLITS responses were more variable, typically exhibiting a smaller ramping phase and seconds-long spikes of activity rather than minutes-long plateaus (Figure 8C)."

This statement does not appear fully consistent with the data in Figure 8. While in figure 8B it looks like GCaMP and mTq-CaFLITS have very similar profiles, these curves come from one single experiment out of a very variable dataset (see Figure 8C). If one would for example choose the second curve of GCaMP in Figure 8C, it would look very similar to the response of G-CaFLITS in figure 8B, and the argument would be reversed. How do the averages look like?

Indeed, the dynamics of the responses are very variable and we do not want to draw attention to these differences in the dynamics, so we have removed the comparison. Instead, the difference in intensity change and lifetime contrast are of importance here. To answer the

question of the reviewer, we have added a new panel (D) which shows the average responses for each of the GECIs.

(7) "Although the calibration is equipment independent under ideal conditions, and only needs to be performed once, we prefer to repeat the calibration for different setups to account for differences in temperature or pulse frequency."

While I generally agree with the statement, it is imprecise. A change in temperature is generally expected to affect the K_d , so rather than "preferring to repeat", it is a requirement for accurate quantification at different concentrations. I am not sure I understand what the pulse frequency is in this context, and how it affects the K_d .

We thank the referee for pointing out that our text is imprecise and confusing. What we meant to say is that we see differences between different set-ups and we have clarified this by changing the text. We have also added that it is “necessary” to repeat the calibration:

“Although the calibration is equipment independent under ideal conditions, and only needs to be performed once, we do see differences between different set-ups. Therefore, it is necessary to repeat the calibration for different set-ups.”

(8) "A recent effort to generate a green emitting lifetime biosensor used a GFP variant as a template (Koveal et al., 2022), and the resulting biosensor was pH sensitive in the physiological range. On the other hand, biosensors with a CFP-like chromophore are largely pH insensitive (van der Linden et al., 2021; Zhong et al., 2024)."

The dismissal of the use of T-Sapphire as a pH independent template is inaccurate. The same group has previously reported other sensors (SweetieTS for glucose and Peredox for redox ratio) that are not pH sensitive. Furthermore, in Koveal et al. also many of the mTq2-based variants showed a pH response, suggesting that the pH dependence for the Lilac sensor might be more complex. Still, G-CaFLITS present advantages in terms of the possibility to excite at longer wavelengths, which could be mentioned instead.

We only want to make the point that adding the T203Y mutation to Turquoise-based lifetime biosensors may be a good approach for generating pH insensitive green biosensors. There is no point in dismissing other green biosensors and we have changed the text to: “Since biosensors with a CFP-like chromophore are largely pH insensitive (van der Linden et al., 2021; Zhong et al., 2024), and we show here that the pH independence is retained for the Green Ca-FLITS, we expect that adding the T203Y mutation to a cyan sensor is a good approach for generating pH-insensitive green lifetime-based sensors.”

(9) "Usually, a higher QY results in a higher intensity; however, in G-Ca-FLITS the open state has a differential shaped excitation spectrum which leads to a decreased intensity. These effects combined have resulted in a sensor where the two different states have a similar intensity despite displaying a large QY and lifetime contrast."

This statement does not seem to reflect the excitation spectra of Figure 1. If this explanation would be true, wouldn't there be an isoemissive point in the excitation spectrum (i.e. an excitation wavelength at which emission intensity would not change)?

The excitation spectra in figure 1 are not ideal for the interpretation as these are not normalized. The normalized spectra are shown in figure S10, but for clarity we show the normalized spectra here below as well. For the FD-FLIM experiments we used a 446 nm LED that excites the calcium bound state more efficiently. Therefore, the lower brightness due to a lower QY of the calcium bound state is compensated by increased excitation. So the limited

change in intensity is excitation wavelength dependent. We have added a sentence to the discussion to stress this:

“The smallest intensity change is obtained when the calcium-bound state is preferably excited (i.e. near 450 nm) and the effect is less pronounced when the probe is excited near its peak at 474 nm”

(10) "We evaluated the use of Tq-Ca-FLITS and G-Ca-FLITS for 2P-FLIM and observed a surprisingly low brightness of the green variant in an intact fly brain. This result is consistent with a study finding that red-shifted fluorescent-protein variants that are much brighter under one-photon excitation are, surprisingly, dimmer than their blue cousins in multi-photon microscopy (Molina et al., 2017). The responses of both probes were in line with their properties in single photon FLIM, but given the low brightness of G-Ca-FLITS under 2-photon excitation, the Tq-Ca-FLITS may be a better choice for 2P-FLIM experiments."

The differences appear strikingly high, and it seems improbable that a reduction in two-photon absorption coefficient might be the sole cause. How can the authors rule out a problem in expression (possibly organism-specific)?

The reviewers are correct that the changes in brightness between G-Ca-FLITS and Tq-Ca-FLITS may arise from changes in expression levels. It is difficult to calibrate for these changes explicitly without a stable reference fluorophore. However, both the G-Ca-FLITS and Tq-Ca-FLITS transgenic flies produced used the same plasmid backbone (the Janelia 20x-UAS-IVS plasmid), landed in the same insertion site (VK00005) of the same genetic background and were crossed to the same Janelia driver line (R60D05-Gal4), so at the level of the transcriptional machinery or genetic regulatory landscape the two lines are probably identical except for the few base pair differences between the G-Ca-FLITS and Tq-Ca-FLITS sequence. But the same level of transcription may not correspond to the same amount of stable protein in the ellipsoid body. So, we cannot rule out any organism-specific problems in expression. To examine the 2P excitation efficiency relative to 1P excitation efficiency, we have measured the fluorescence intensity of purified G-Ca-FLITS and Tq-Ca-FLITS on beads. See also response to reviewer 3 and supplemental figure S14

Suggestions

(1) The underlying assumption of any experiment using a biosensor is that the concentration of the biosensor should be roughly 2 orders of magnitude lower than the concentration of the analyte, otherwise the calibration equations do not hold. When measuring nM concentrations of calcium, this problem can be in principle very significant, as the concentration of the sensor in cells is likely in the low micromolar range. Calcium regulation by the cell should compensate for the problem, and the equations should hold. However, this might not hold true during experimental conditions that would disrupt this tight regulation. It might be a good thing to add a sentence to inform users about the limitations in interpreting calcium concentration data under such conditions.

Good point. We have added this to the discussion: “All calcium indicators also act as buffers, and this limits the accuracy of the absolute measurements, especially for the lower calcium concentrations (Rose et al., 2014), as the expression of the biosensor is usually in the low micromolar range.”

(2) Different methods of lifetime "averaging", such as intensity or amplitude-weighted lifetime in time domain FLIM or phase and modulation in frequency domain might lead to different Kd in the same calibration experiment. This is an underappreciated factor

that might lead to errors by users. Since the authors conducted calibrations using both frequency and time-domain, it would be useful to mention this fact and maybe add a table in the Supporting Information with the minima, maxima and Kds calculated using different lifetime averaging methods.

To avoid biases due to fitting we prefer to use the phasor plot, this can be used for both frequency and time-domain methods and we added a sentence to the discussion to highlight this: “We prefer to use the phasor analysis (which can be used for both frequency- and time-domain FLIM), as it makes no assumptions about the underlying decay kinetics.”

(3) The origin of the redshift observed in G-CaFLITS is likely pi-stacking, similar to the EGFP-to-EYFP case. While previous studies suggest that for mTq2 based sensors a change in rigidity would lead to a change in the non-radiative rate, which would result in similar changes in quantum yield and (amplitude-weighted average) lifetime. If pi-stacking plays a role, there could be an additional change in the radiative rate (as suggested also by the change in absorption spectra). Could this play a role in the relation between brightness and lifetime in G-CaFLITS? Given the extensive data collected by the authors, it should be possible to comment on these mechanistical aspects, which would be useful to guide future design.

We do appreciate this suggestion, but we currently do not have the data to answer this question. The inverted response that we observe, solely due to the introduction of the tyrosine is puzzling. Perhaps introduction of the mutation that causes the redshift in other cyan probes will provide more insight.

Reviewer #2 (Recommendations for the authors):

Specific points:

The first section of Results is basically a description of how they chose the lysis conditions for screening in bacteria. I didn't see anything particularly novel or interesting about this, anyone working with protein expression in bacteria likely needs to optimize growth, lysis, purification, etc. This section should be moved to the Methods.

As reviewer 1 lists the thorough documentation of this approach as one of the strengths, we prefer to keep it like this. We see this section as method development, rather than purely a method. When this section would be moved to methods, it remains largely invisible and we think that's a shame. Readers that are not interested can easily skip this section.

In the Results section Characterization of G-Ca-FLITS, the authors state "Here, the calcium affinity was $K_D = 339$ nM, higher compared to the calibration at 37°C. This is in line with the notion that binding strength generally increases with decreasing temperature." However, the opposite appears to be true - at 37°C they measured a K_D of 209 nM which would represent higher binding strength at higher temperature.

Thanks for catching this, we've made a mistake. We rephrase this to “higher compared to the calibration at 37 °C. This is unexpected as it not in line with the notion that binding strength generally increases with decreasing temperature.”

In Figure 8c, there should be a visual indicator showing the onset of application of high potassium, as there is in 8b.

This is a good suggestion; a grey box is added to indicates time when high K⁺ saline was perfused.

Reviewer #3 (Recommendations for the authors):

I think the science of the manuscript is sound and the presentation is logical and clear. I have some stylistic recommendations.

Supp Fig 1: The figure requires a bit of "eyeballing" to decide which conditions are best, and figuring out which spectra matched the final conditions took a little effort. Is there a way to quantify the fluorescence yield to better show why the one set of conditions was chosen? If it was subjective, then at least highlight the final conditions with a box around the spectra, making it a different colour, or something to make it stand out.

Thanks for the comment; we added a green box.

Supp Fig 3: Similar suggestion. Highlight the final variant that was carried forward (T203Y). The subtle differences in spectra are hard to discern when they are presented separately. How would it look if they were plotted all on one graph? Or if each mutant were presented as a point on a graph of Peak Em vs Peak Ex? Would T203Y be in the top right?

We have added a light blue box for reference to make the differences clearer.

Supp Fig 4 & Fig 1: Too much of the graph show the uninteresting tails of the spectra and condenses the interesting part. Plotting from 400 nm to 600 nm would be more informative.

We appreciate the suggestion but disagree. We prefer to show the spectra in its entirety, including the tails. The data will be available so other plots can be made by anyone.

Fig 3a: People who are not experts in lifetime analysis are probably not very familiar with the phase/modulation polar plot. There should be an additional sentence or two in the main text that briefly describes the basis for making the polar plot and the transformation to the fractional saturation plot in 3B. I can't think of a good way to transform Eq 3 from Supp Info into a sentence, but that's what I think is needed to make this transformation clearer.

We appreciate the suggestion and feel that it is well explained here:

"The two extreme values (zero calcium and 39 μ M free calcium) are located on different coordinates in the polar plot and all intermediate concentrations are located on a straight line between these two extremes. Based on the position in the polar plot, we determined the fraction of sensor in the calcium-bound state, while considering the intensity contribution of both states"

Fig 4: The figure is great, and I love the comparison of different calcium sensors. But where is Tq-Ca-FLITS? I get that this is a figure of green calcium sensors, but it would be nice to see Tq-Ca-FLITS in there as well. The G-Ca-FLITS is compared to Tq-Ca-FLITS in Fig 5. Maybe I'm just missing why the bottom panel of Fig 5 cannot be replotted and included in Fig 4.

The point is that we compare all the data with identical filter sets, i.e. for green FPs. using these ex/em settings, the Tq probe would seriously underperform. Note that the data in fig. 5 is not normalized to a reference RFP and can therefore not be compared with data presented in figure 4.

Fig 6: The BOEC data could easily be moved to Supp Figs. It doesn't contribute much relevant info.

We are not keen of moving data to supplemental, as too often the supplemental data is ignored. Moreover, we think that the BOEC data is valuable (as BOEC are primary cells and therefore a good model of a healthy human cell) and deserves a place in the main manuscript.

2P FLIM / Fig 8 / Fig S4: The lack of brightness of G-Ca-FLITS in the 2P FLIM of fruit fly brain could have been predicted with a 2P cross section of the purified protein. If the equipment to perform such measurements is available, it could be incorporated into Fig S4.

Unfortunately, we do not have access to equipment that measures the 2P cross section. As an alternative, we compared the 2P excitation efficiency with 1P excitation efficiency. To this end, we have used beads that were loaded with purified G-Ca-FLITS or Tq-Ca-FLITS. We have evaluated the fluorescence intensity of the beads using 1P (460 nm) and 2P (920 nm) excitation. Although the absolute intensity cannot be compared (the G-Ca-FLITS beads have a lower protein concentration), we can compare the relative intensities when changing from 1P to 2P. The 2P excitation efficiency of G-Ca-FLITS is comparable (if not better) to that of Tq-Ca-FLITS. This excludes the option that the G-Ca-FLITS has poor 2P excitability. We will include this data as figure S12.

We also have added text to the results: “We evaluated the relative brightness of purified Tq-Ca-FLITS and G-Ca-FLITS on beads by either 1-Photon Excitation (1PE) (at 460 nm) or 2-Photon Excitation (2PE) (at 920 nm) and observed a similar brightness between the two modes of excitations (figure S14). This shows that the two probes have similar efficiencies in 2PE and suggest that the low brightness of G-Ca-FLITS in *Drosophila* is due to lower expression or poor folding.” and discussion: “The responses of both probes were in line with their properties in single photon FLIM, but given the low brightness of G-Ca-FLITS under 2-photon excitation in *Drosophila*, the Tq-Ca-FLITS is a better choice in this system. Yet, the brightness of G-Ca-FLITS with 2PE at 920 nm is comparable to Tq-Ca-FLITS, so we expect that 2P-FLIM with G-Ca-FLITS is possible in tissues that express it well.”

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